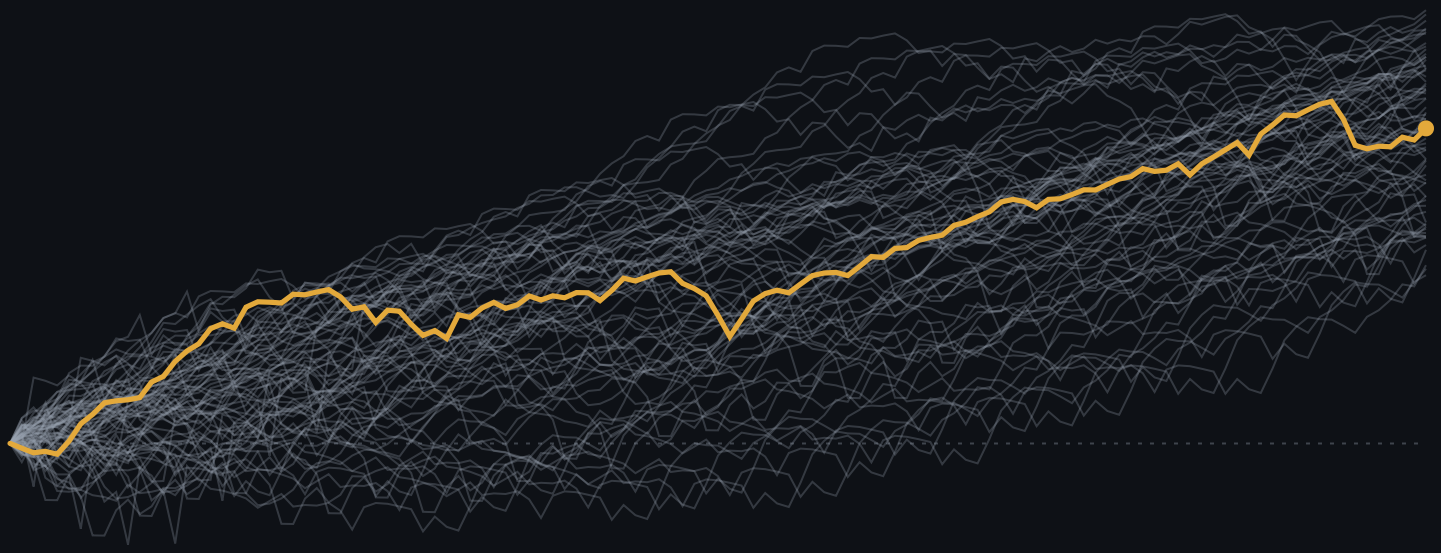
The complete retirement planning toolkit.

A calculator, two guided walkthroughs and fourteen planning tools, each checked against every market since 1926. Free, and you don't need an account.



Every 30-year stretch since 1926 for a 60/40 portfolio, after inflation. The amber line is the middle one.

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Calculators · Retirement income · Saving and investing · Everyday money · Guided walkthroughs · Putting it together

PART 1

The overview.

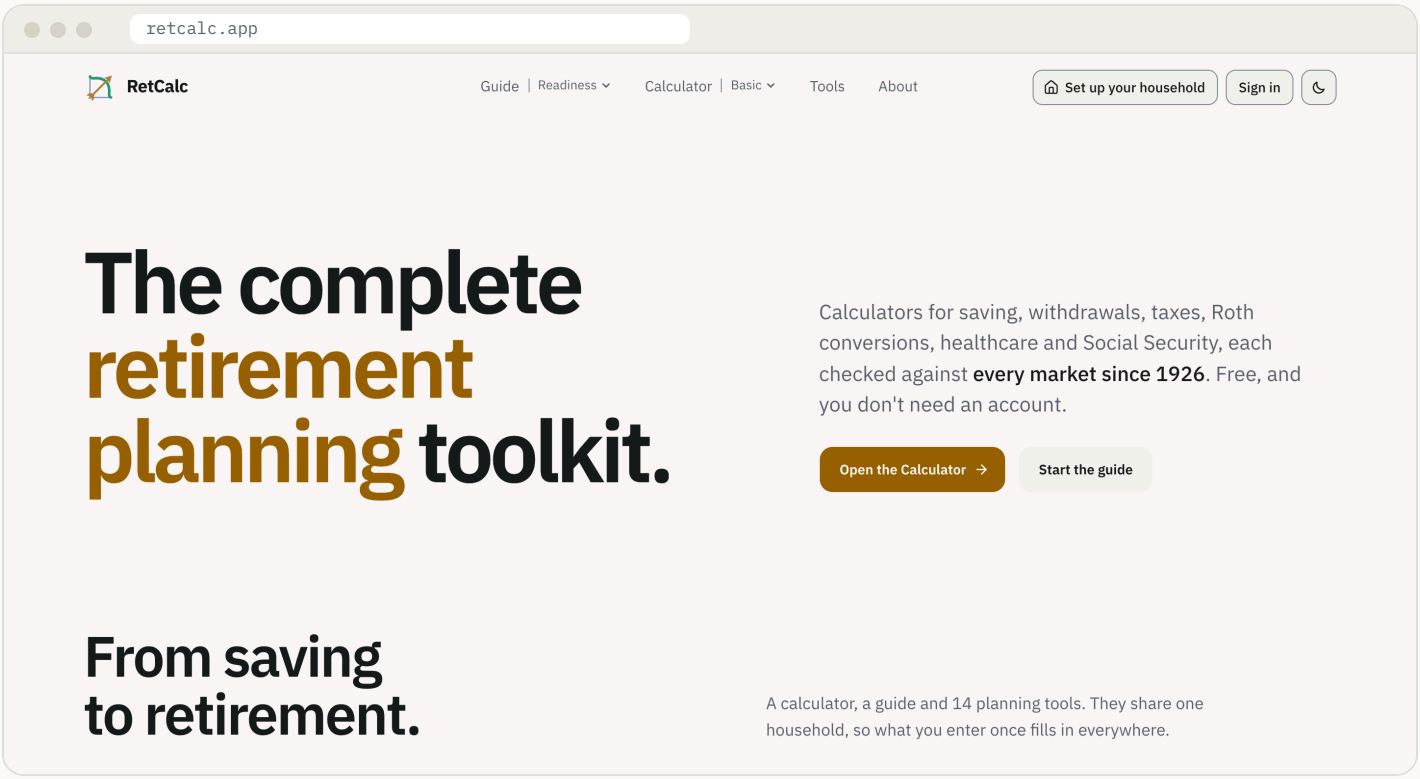
Free, private, and tested against a hundred years of real markets.

What RetCalc is	04	Highlights	07
How it works	05	Every tool at a glance	08
A tool for every stage	06		

WHAT IT IS

A whole retirement plan, worked out in your own browser.

RetCalc is a free retirement planning toolkit at **retcalc.app**. It takes someone from their first paycheck to their last withdrawal: what to save, where to put it, whether it will last, what it will cost in tax and health insurance, and when to claim Social Security. Every projection is tested against real market history, month by month, back to 1926.



14 planning tools, in three groups	**3** calculators: Basic, Advanced and Stages	**2** guided walkthroughs that end in a printable plan	**100** years of monthly market history, 1926 to 2025	**$0** no ads, no paywall, no account needed

For someone starting out

What a paycheck really takes home, a budget that adds up, and how today's saving grows into a retirement income.

For someone getting close

When to claim, how to cover the years before 59½ and 65, whether to convert to Roth, and what the best plan looks like.

For someone already retired

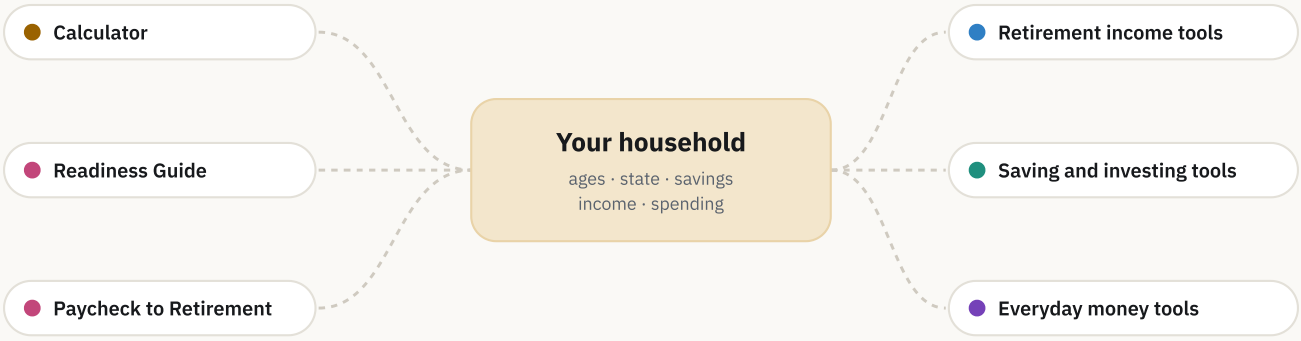
How much is safe to spend, which withdrawal rule to follow, and how the plan would have fared in the worst markets on record.

HOW IT WORKS

Three ideas run through every page.

1 One household, every tool.

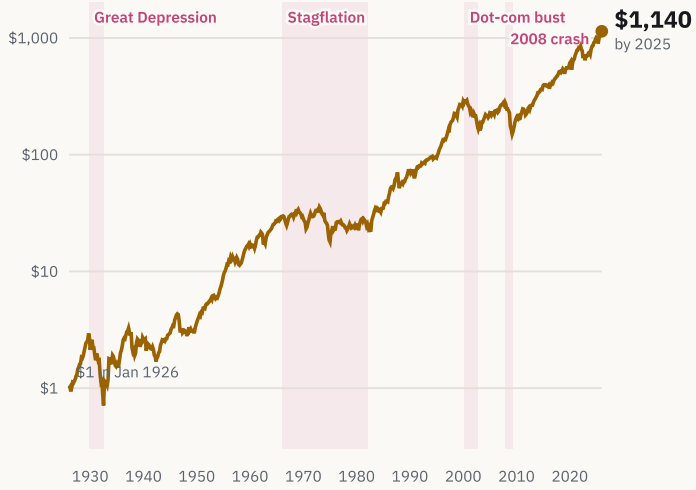
Ages, state, savings, income and spending are entered once and fill in everywhere. Results also carry forward: the Calculator hands its balance to the Drawdown Simulator, Income Tax hands its paycheck to the Budget, and the Budget's savings split goes to the Advanced or Stages calculator.



2 Tested against history, not one average.

A single average return hides the order returns come in, and that order decides whether a retirement lasts. RetCalc replays every real starting month since January 1926: the Depression, 1970s stagflation, 2000 and 2008. Where history runs out, it runs 5,000 Monte Carlo simulations per redraw.

One dollar in US stocks in January 1926 grew to about **$1,140** after inflation by the end of 2025, in the same data the tools use. The shaded years are when it didn't feel that way.



US stocks after inflation, dividends reinvested. Log scale. Shiller (Yale) and BLS CPI-U.

3 Private by default.

Every calculation runs in the visitor's own browser, and nothing they type is sent anywhere. An account is optional: signing in with a one-time email code or Google simply keeps a household, saved scenarios and Guide answers in step across devices. As the site puts it, *"Nothing leaves your browser unless you sign in to save it."*

WHERE TO START

A tool for every stage of the road.

The tools follow the order life brings the questions. Most people start with the Calculator or one of the two guided walkthroughs, which open the right tools along the way.

1 2 3 4

Today

Get the basics right

- Income Tax
- Budget
- Debt Payoff
- Mortgage Calculator
- Rent vs. Buy

Saving years

Grow it

- Calculator
  Basic · Advanced · Stages
- FIRE Calculator
- Portfolio Backtest
- College Savings

Getting close

Plan the handoff

- Social Security Calculator
- Healthcare Cost Planner
- Early Retirement Bridge
- Roth Conversion Calculator
- Retirement Plan Optimizer

Retired

Make it last

- Drawdown Simulator
- Roth Conversion Calculator
- Healthcare Cost Planner

GUIDED WALKTHROUGH

Retirement Readiness Guide

A check-up in twelve short cards (about ten minutes) or a full walkthrough. It finds the number to aim for, tests it against history, scores readiness out of 100, and ends in a full-color report of about a dozen pages.

GUIDED WALKTHROUGH

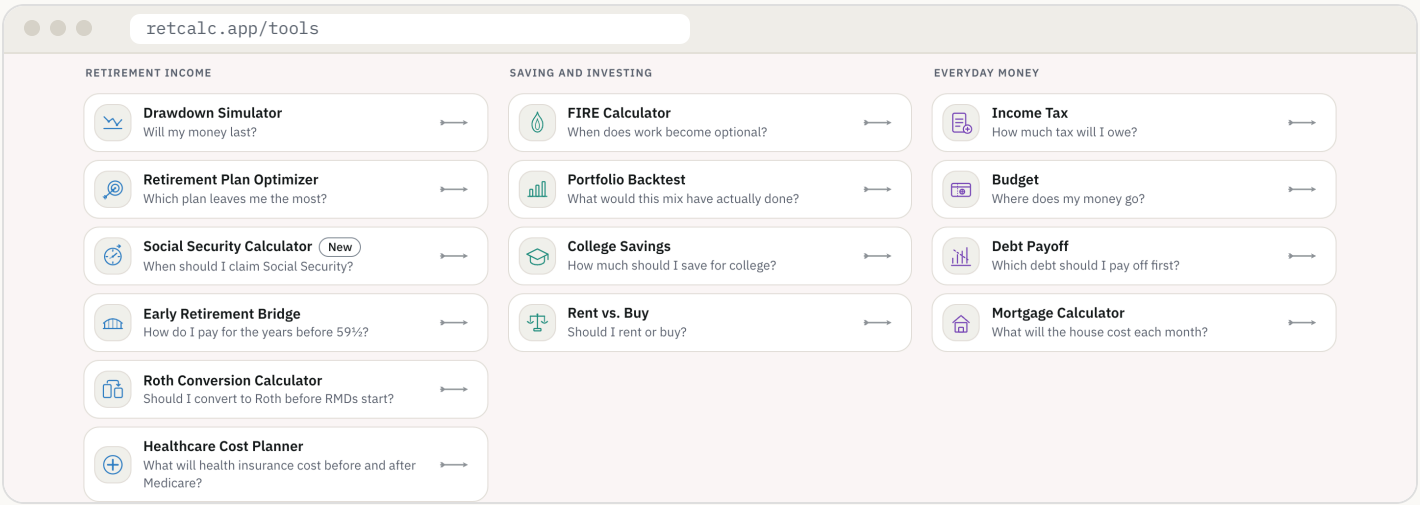
Paycheck to Retirement

Five steps, about fifteen minutes: from gross pay to take-home, a budget, where each paycheck goes (401(k), HSA, IRA, brokerage), and where that saving leads by retirement, children included.

CALCULATOR

Basic, Advanced and Stages

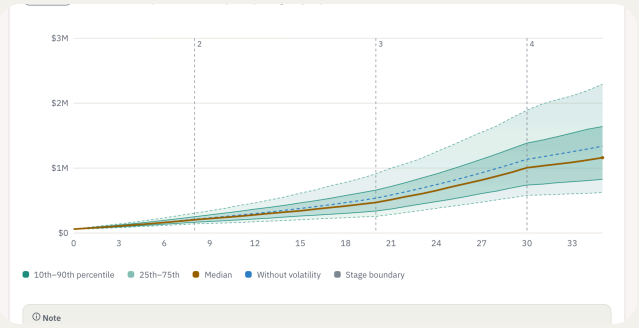
Six questions for a quick answer; then fees, taxes, account types and a target; then a plan in stages, each with its own saving, return and length.



The tool list, compact view. All fourteen tools in their three groups, each with the question it answers.

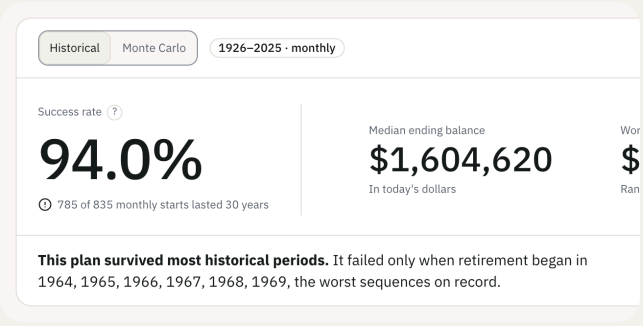
HIGHLIGHTS

Six places to see what it can do.



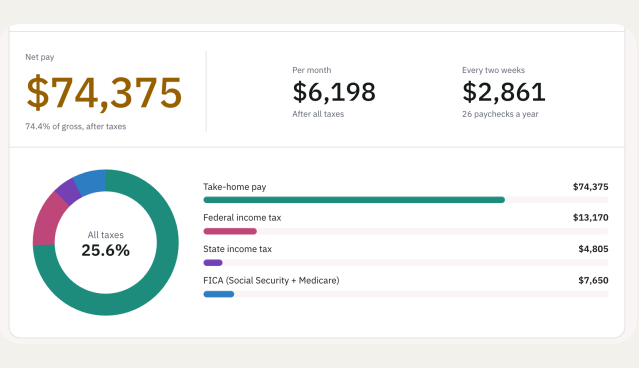
Stages calculator

A savings plan in stretches: early career, kids at home, peak earning, coasting. Each stage has its own saving, return and volatility, chained into one projection and tested three ways.



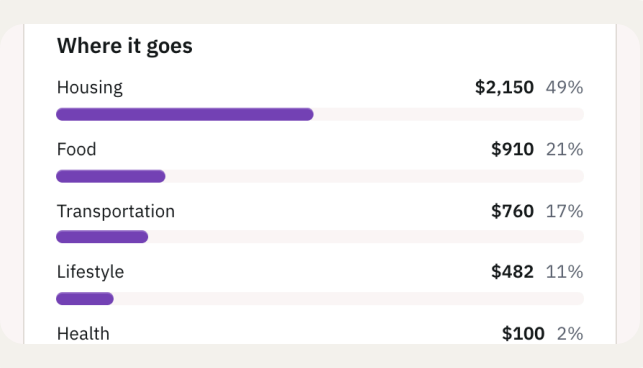
Drawdown Simulator

Fifteen withdrawal strategies replayed through every retirement since 1926, then ranked against each other at the same level of risk.



Income Tax

2026 federal, state (all 50 plus DC) and FICA tax on a salary, or a retirement mode that stacks withdrawals, gains and Social Security the way the return really does.



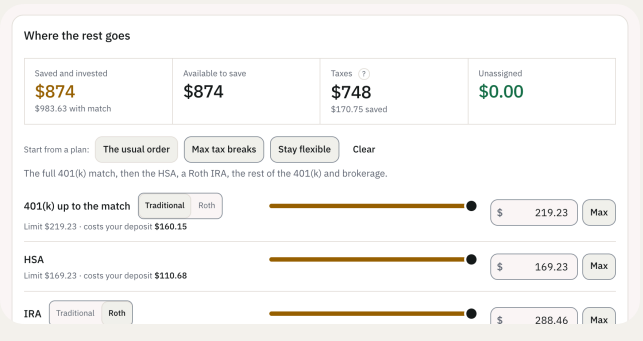
Budget and Paycheck

A budget, then a paycheck that pays it: taxes rerun live as sliders move money into the 401(k) match, HSA, IRA and brokerage, in the order you choose.



Retirement Readiness Guide

A check-up that scores readiness out of 100, tests the plan against history and prints a report of about a dozen pages.



Paycheck to Retirement

Gross pay to retirement in five steps, running the real tools inside it and copying the results into them at the end.

AT A GLANCE

Every tool, and the question it answers.

CALCULATORS

Calculator: Basic	What will my savings grow to?
Calculator: Advanced	Will I reach my target, after fees and tax?
Calculator: Stages	What if my saving changes over the years?

RETIREMENT INCOME

Drawdown Simulator	Will my money last?
Retirement Plan Optimizer	Which plan leaves me the most?
Social Security Calculator	When should I claim Social Security?
Early Retirement Bridge	How do I pay for the years before 59½?
Roth Conversion Calculator	Should I convert to Roth before RMDs start?
Healthcare Cost Planner	What will health insurance cost before and after Medicare?

SAVING AND INVESTING

FIRE Calculator	When does work become optional?
Portfolio Backtest	What would this mix have actually done?
College Savings	How much should I save for college?
Rent vs. Buy	Should I rent or buy?

EVERYDAY MONEY

Income Tax	How much tax will I owe?
Budget	Where does my money go?
Debt Payoff	Which debt should I pay off first?
Mortgage Calculator	What will the house cost each month?

GUIDED WALKTHROUGHS

Retirement Readiness Guide	How ready am I, and what should I do next?
Paycheck to Retirement	Where should each paycheck go?

ALSO ON THE SITE

29 plain-English articles, each with worked examples the test suite checks against the calculator, so the numbers in the writing always match the numbers in the tools.

PART 2

The deep dive.

Every tool on the site, what it asks for and what it gives back.

Basic and Advanced	10
Stages calculator	11
Drawdown Simulator	12
Retirement Plan Optimizer	13
Social Security Calculator	14
Early Retirement Bridge	15
Roth Conversion Calculator	16
Healthcare Cost Planner	17
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Mortgage Calculator	25
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From claiming to drawdown	30

Basic and Advanced

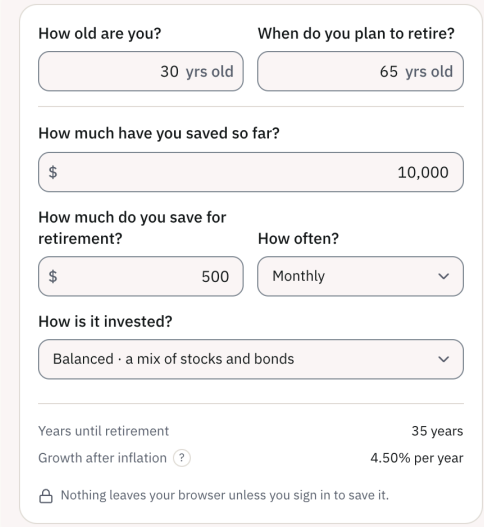
What will my savings grow to, and will it be enough?

The Calculator comes in three depths. Basic answers in six questions; Advanced adds what a real plan has to carry: fees, inflation, account types, an employer match, 2026 taxes and a target to reach.

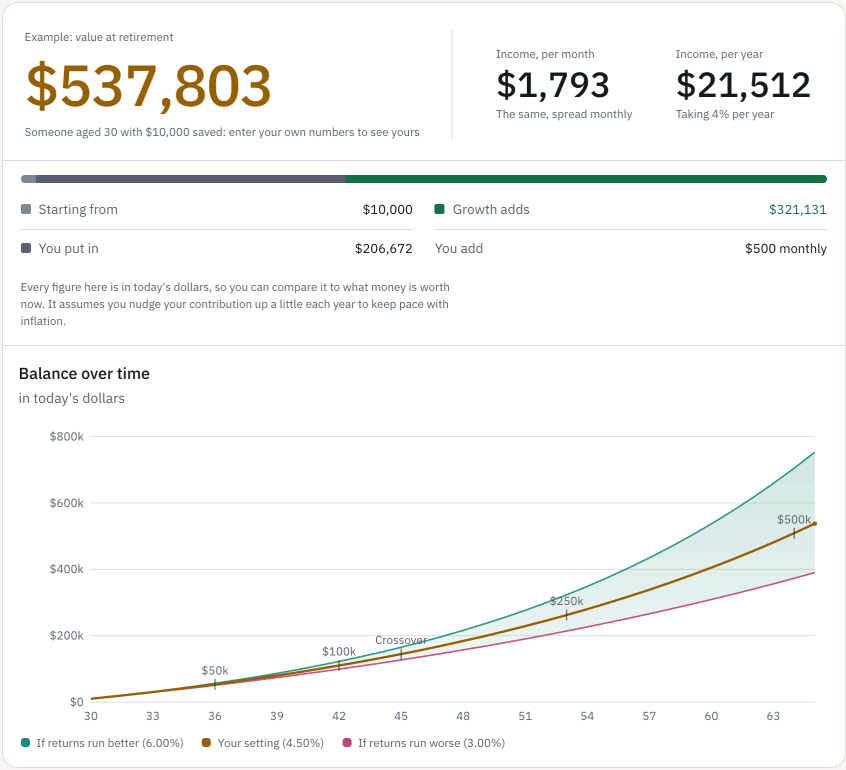
B · A

retcalc.app/calculator

BASIC · SIX QUESTIONS



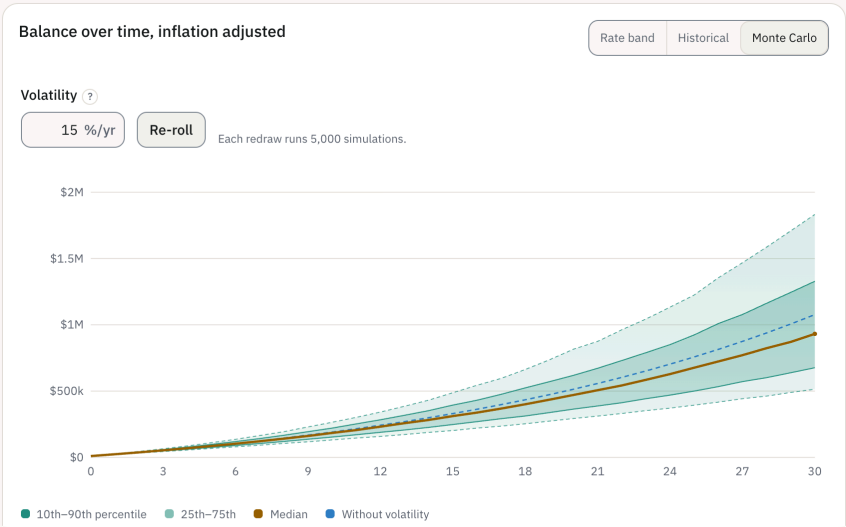
Age, retirement age, savings, how much and how often, and one of five risk levels. Out comes the balance at retirement in today's dollars and the income it could pay.



Basic. The answer, what built it, and a band for better and worse returns with milestones along the line.

ADVANCED · THE WHOLE PLAN

- Traditional, Roth, brokerage and HSA balances, each with its own contribution
- Employer match, contribution growth, fees and a glide path
- Filing status and state set the tax in retirement
- Solve for an after-tax income or a portfolio value
- **Reach your target:** the contribution or the years it would take, applied in one click
- Chart three ways: a rate band, every historical window since 1926, or 5,000 Monte Carlo runs



Advanced, Monte Carlo. Percentile bands across 5,000 runs, against the plan without volatility.

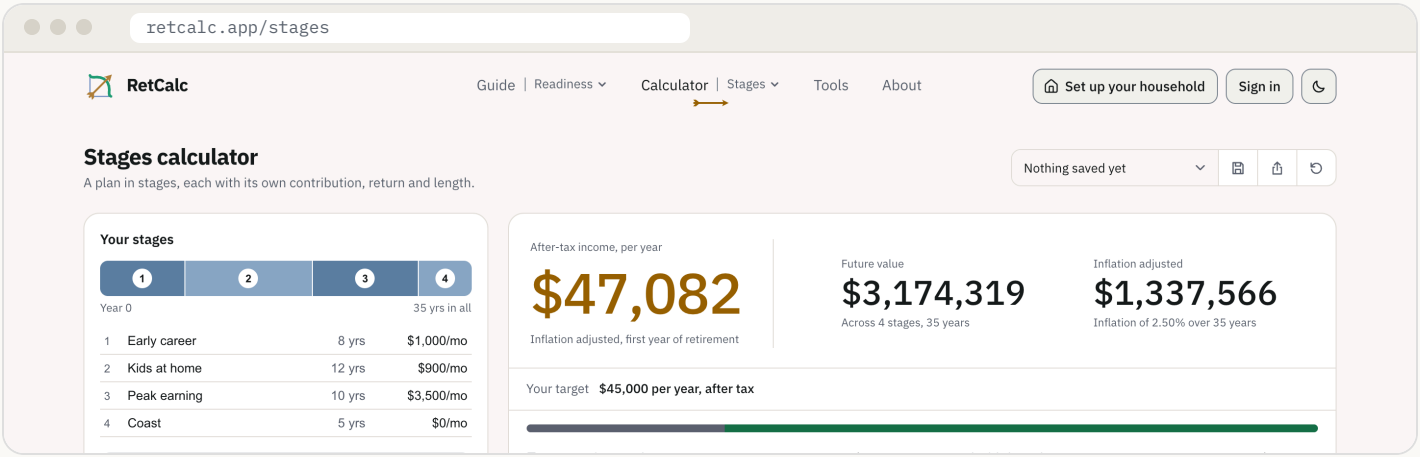
Stages calculator

What if my saving changes over the years?

Real plans come in stretches. Stages chains them into one projection: each stage has its own length, contribution, raises, return, volatility, glide path and account split.

S

retcalc.app/stages



What makes it different

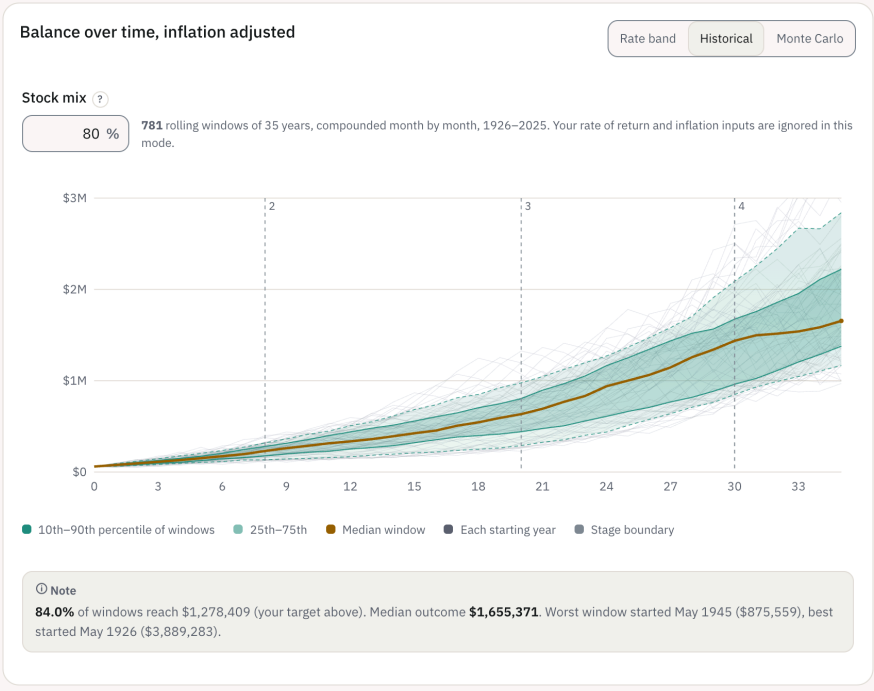
- Any number of named stages, sized by their years on a stage bar
- A **Planning for kids** builder rebuilds the stages around the children you have or plan
- Stage boundaries drawn right on the chart
- Solves the final stage for you: the contribution, or the length, that reaches the target

Tested three ways

A ±rate band, every rolling window since 1926 with a stock mix that can glide, and 5,000 Monte Carlo runs with volatility set per stage.

Hands off to

The Drawdown Simulator, to see how long the balance lasts once spending starts.



Sample plan, every historical window since 1926. Early career, kids at home, peak earning, then coasting. Dashed lines mark each new stage.

Stage by stage CSV

Stage	Years	Return	Start balance	Contributions	Growth	End balance
1	8	7.90%	$60,000	$110,571	$89,029	$259,600
2	12	7.90%	$259,600	$144,851	$473,733	$878,183
3	10	6.90%	$878,183	$481,483	$1,023,590	$2,383,257
4	5	5.90%	$2,383,257	$0	$791,063	$3,174,319

Drawdown Simulator

Will my money last?

Most calculators ask how much you will have. This one asks the harder question: once you start spending it, will it last? It replays a plan through every real retirement since 1926, month by month, with Social Security, pensions, annuities and one-time costs.

01

retcalc.app/drawdown

Historical | Monte Carlo | 1926–2025 · monthly | Pin as baseline

Success rate

94.0%

785 of 835 monthly starts lasted 30 years

Median ending balance

$1,604,620

In today's dollars

Worst case

$0

Ran out in 50 of 835 retirements

This plan survived most historical periods. It failed only when retirement began in 1964, 1965, 1966, 1967, 1968, 1969, the worst sequences on record.

15 STRATEGIES, FIVE FAMILIES

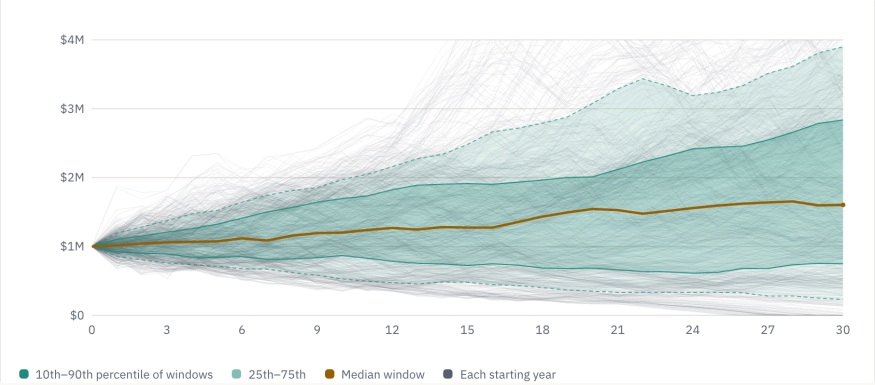
Steady income
Fixed, rising with inflation (4% rule) · Kitces ratchet

Share of portfolio
Fixed percentage · 95% rule (Clyatt) · 1/N · RMD method · Variable percentage (VPW)

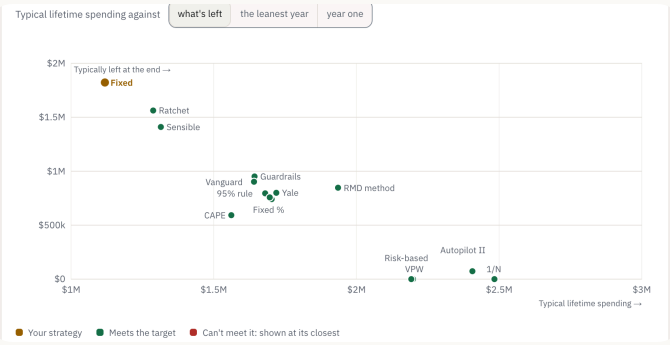
Guardrails
Guyton-Klinger · Risk-based guardrails

Smoothed
Floor and ceiling · Vanguard dynamic spending · Yale endowment · Hebeler Autopilot II · Sensible withdrawals

Valuation
CAPE-based



Every historical starting month. Percentile bands across 835 real 30-year retirements, each start drawn in gray.



Strategy showdown. All 15 tuned to the same risk, then compared on lifetime spending and what is left.

Rate \ Stocks	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
2.50%	under	under	under	under	under	under	under	under	under	under	under
2.75%	under	under	under	under	under	under	under	under	under	under	under
3.00%	under	under	under	under	under	under	under	under	under	under	under
3.25%	✕ 69	94	✓ 100	✓ 100	✓ 100	✓ 100	✓ 100	✓ 100	✓ 100	✓ 100	100
3.50%	✕ 51	✕ 75	96	100	✓ 100	✓ 100	✓ 100	✓ 100	✓ 100	100	99
3.75%	✕ 47	✕ 60	✕ 81	95	97	99	100	✓ 100	✓ 100	100	99
4.00%	✕ 42	✕ 49	✕ 68	✕ 85	91	94	**94**	95	96	96	95
4.25%	✕ 39	✕ 43	✕ 56	✕ 72	✕ 86	✕ 89	91	92	93	92	91
4.50%	✕ 37	✕ 39	✕ 52	✕ 63	✕ 74	✕ 83	✕ 86	✕ 87	✕ 88	✕ 88	✕ 88
4.75%	✕ 36	✕ 37	✕ 43	✕ 56	✕ 67	✕ 75	✕ 81	✕ 83	✕ 83	✕ 84	✕ 84
5.00%	✕ 29	✕ 36	✕ 39	✕ 52	✕ 62	✕ 70	✕ 75	✕ 79	✕ 80	✕ 80	✕ 80
5.25%	✕ 25	✕ 31	✕ 36	✕ 46	✕ 57	✕ 65	✕ 70	✕ 73	✕ 76	✕ 77	✕ 78
5.50%	✕ 23	✕ 28	✕ 32	✕ 39	✕ 51	✕ 59	✕ 66	✕ 69	✕ 72	✕ 74	✕ 75

Success grid. Withdrawal rate against stock share. Click a cell to use it.

Retirement Plan Optimizer

Which plan leaves me the most?

Thousands of combinations of Social Security claiming ages, withdrawal order, Roth conversions and income guards, each run through every market since 1926, to find the plan that leaves the most, lasts the longest or spends the most.

02

retcalc.app/optimizer

Left for you and your heirs after tax, in a typical market

$4.01M

+$375k vs. $3.63M the usual way

Lifetime tax	Lasted in	Social Security at
$167,402	100%	70 & 70
was $340,554 ↓ better	was 100%	was 67 & 67

Tried **every one of 4,941 plans** in all **66 historical retirements** since 1926: 326,106 retirements simulated.

How it searches

Every pair of claiming ages from 62 to 70 for a couple (81 pairs), crossed with how far to fill each bracket with conversions and whether to stay under the IRMAA and ACA income lines. In the sample above that came to **4,941 plans** in **66 historical retirements**: 326,106 retirements simulated.

Three goals

- Leave the most
- Make it last
- Spend the most

What you get

- How much each kind of change adds
- A year-by-year roadmap in plain words
- Tax and premiums against the usual way
- Close runners-up, so the choice isn't a black box

What makes the difference

median left after tax

Claim Social Security at 70 for both of you, instead of 67 for both of you +$207k left

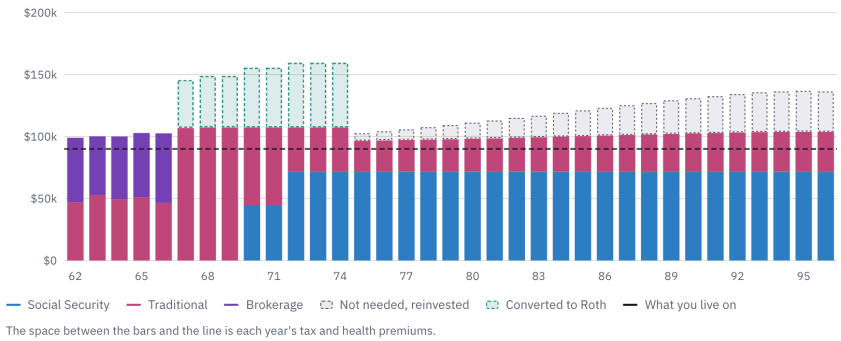
Every year you wait past 67 adds 8% to the check, for life, and it rises with inflation. Your savings carry the years in between, which usually costs less than the bigger check pays back over a long retirement.

Draw traditional money first, up to the top of the 12% bracket, and convert what you don't spend to Roth until 74, and keep income under the ACA subsidy cliff before 65 +$168k left

Traditional money is taxed whenever it comes out. Taking it in the lower-income years, at 12%, beats taking it later, when required distributions and Social Security stack up and push it into higher brackets. Roth money then grows tax-free for you and your heirs.

Where each year's money comes from

your roadmap, average path



The space between the bars and the line is each year's tax and health premiums.

Where each year's money comes from. Social Security, traditional, brokerage, and what is converted to Roth.

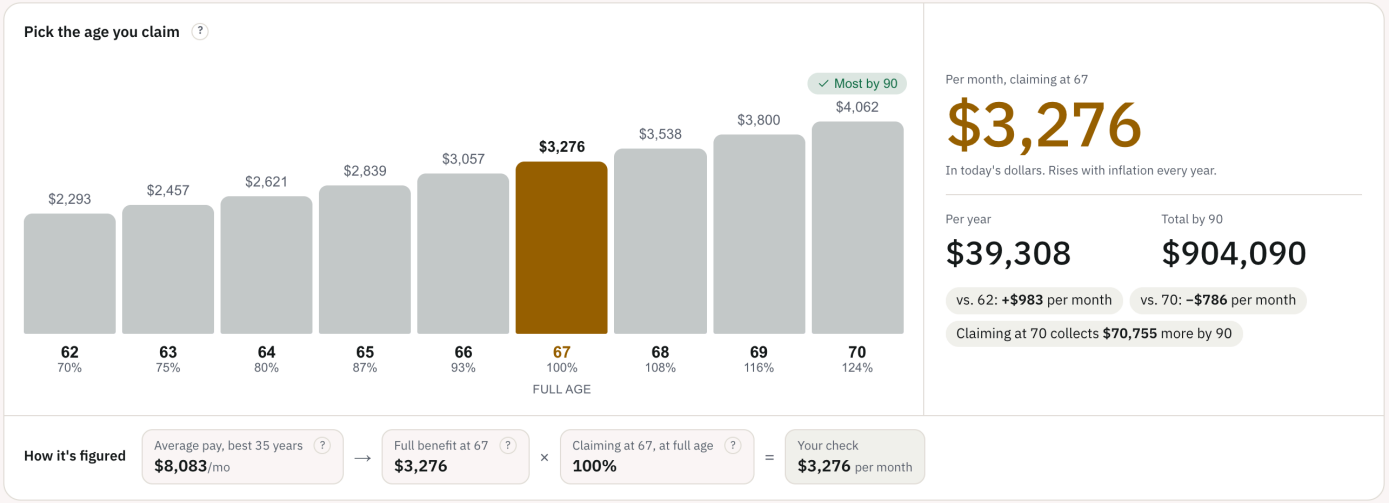
Social Security Calculator

When should I claim Social Security?

What the check will be at every age from 62 to 70, how it is figured, and which age pays off best depending on how long you live. Written as a sentence to fill in; works for a single person or a couple.

03

retcalc.app/social-security

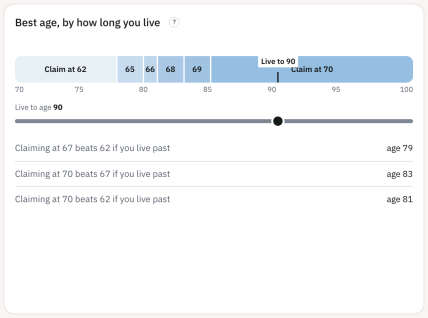


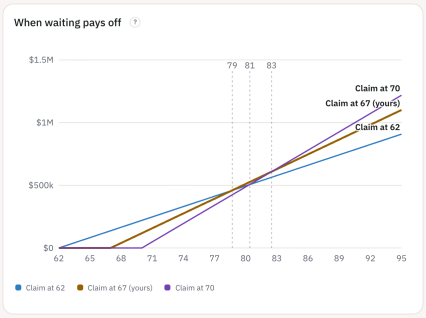
You enter

- Year born and pay, or pay year by year
- A spouse, if there is one
- Life expectancy
- Other retirement income, to work out how much of the benefit is taxed

You get

- A staircase of checks by claiming age: click a step to pick it
- Best age by lifespan, and the break-even
- Survivor benefit and every pair of ages for couples
- Year-by-year and side-by-side tables





Best age, by how long you live; and when waiting pays off.

Every age, side by side

CSV

Claim at	Per month	% of full benefit	Total by 90	Compared to 70
62	$2,293	70%	$770,442	−$204,403
63	$2,457	75%	$795,992	−$178,853
64	$2,621	80%	$817,612	−$157,233
65	$2,839	87%	$851,679	−$123,166
66	$3,057	93%	$880,505	−$94,340
67 full age	**$3,276**	**100%**	**$904,090**	**−$70,755**
68	$3,538	108%	$933,964	−$40,881

Year by year

CSV

Your age	Your benefit	Added up
67	$39,308	$39,308
68	$39,308	$78,617
69	$39,308	$117,925
70	$39,308	$157,233
71	$39,308	$196,541
72	$39,308	$235,850
73	$39,308	$275,158

Every age, side by side; and year by year. Sample: born 1990, earning $85,000.

Early Retirement Bridge

How do I pay for the years before 59½?

Retiring before 59½? It compares a Roth conversion ladder, 72(t) payments, the rule of 55, brokerage money and part-time work, with the tax, penalties and ACA premiums each costs, and shows what is left at 59½.

04

retcalc.app/bridge

Ways to 59½

CSV

Plan	Holds to 59½	Tax	Penalties	Health premiums	At 59½
✓ **Blended plan** Best 72(t) of $60,312/yr · Roth before brokerage	✓ 100%	$22,263	$0	$52,499	$1,936,129
Roth conversion ladder Convert every year; spend each conversion five years later	85%	$23,566	$0	$118,621	$1,814,081
Brokerage, then Roth contributions No conversions; sell taxable investments first	✗ 53%	$13,234	$0	$15,497	$2,002,631
72(t) payments $60,312 per year from an IRA, locked in through age 59	✓ 100%	$22,927	$0	$61,290	$1,922,375
Rule of 55 Needs you to leave your job at 55 or later					–
Pay the 10% penalty Withdraw from the 401(k) or IRA as needed and pay the extra 10%	100%	$54,531	$81,776	$81,449	$1,736,040

In Holds to 59½, ✓ marks 95% or more and ✗ under 80%; between, no mark. Pay the 10% penalty isn't marked: for it only running short counts.

Success rates come from every start year since 1926. Tax, penalties, premiums and the balance at 59½ are the steady path at the long-run average

Routes compared

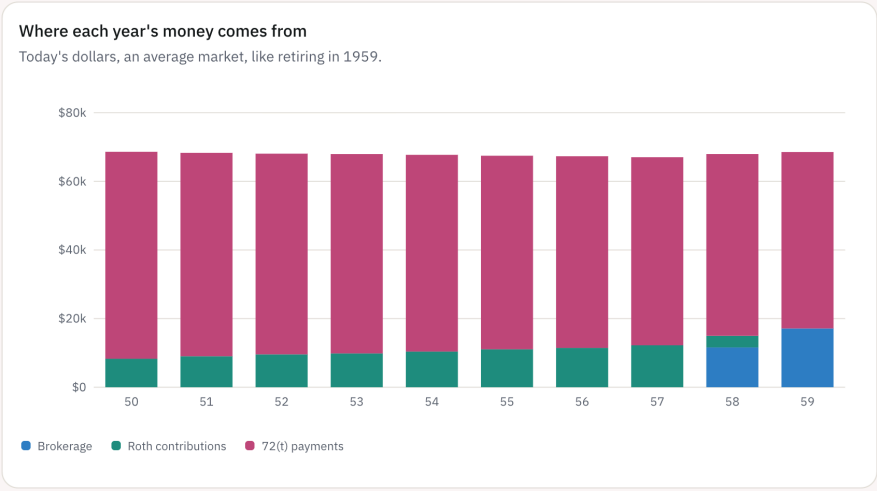
- Roth conversion ladder
- 72(t) / SEPP payments
- Brokerage, then Roth contributions
- Rule of 55
- Part-time work
- Paying the 10% penalty, as a benchmark

Seven conversion fills

From none, through the standard deduction and the tops of the 10% and 12% brackets, to just under the ACA subsidy cliff.

Tested against

Every start since 1926, or Monte Carlo.



Where each year's money comes from, from retirement to 59½.

Roth Conversion Calculator

Should I convert to Roth before RMDs start?

Projects required minimum distributions to age 100, then tests a conversion schedule against doing nothing: lifetime tax, after-tax net worth, IRMAA surcharges and the survivor's higher bracket.

05

retcalc.app/roth

Converting wins on both measures.

Lifetime tax saved
$6,839
Lifetime tax + IRMAA, discounted at 3.0% real

After-tax net worth vs doing nothing
+$223,304
↑ After-tax, at age 92

Peak RMD, doing nothing
$169,099
Converting trims it to $0

Lifetime tax is what you and your heirs hand over; after-tax net worth is what is left standing at age 92, with traditional dollars discounted at 32% because they are still owed to the IRS.

RMDs begin	First RMD, converting	Years converting	Average conversion
Age 75	None in this window	9 years	$143,150/yr

The survivor's bracket. Filing switches to single at age 77. The same income now meets single brackets and a single standard deduction: the marginal rate stays at 10.0%, and tax that year stays at $0.

Conversion strategies

- Fill to a tax bracket
- Fill to an IRMAA threshold
- A fixed amount
- A percentage of the balance
- None, for comparison

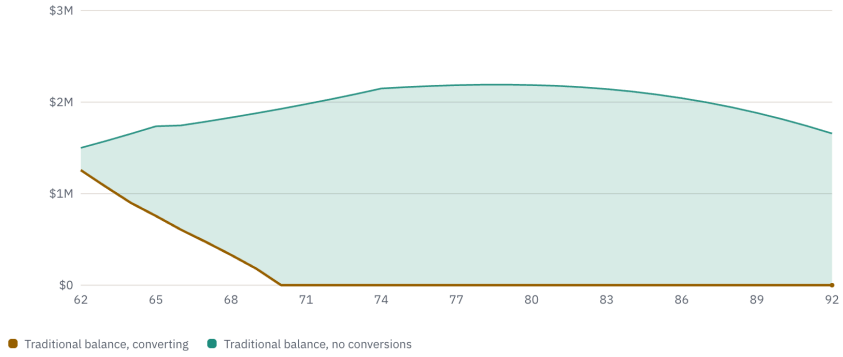
It accounts for

- Social Security and its taxation
- Tax paid from the taxable account
- The survivor filing single after one spouse dies
- Heirs' tax rate on what is left
- A discount rate for present value

Traditional balance

In today's dollars, by age.

Balance | Tax



Traditional balance, converting against doing nothing.

Converting vs. not

Lifetime income tax, converting	$305,186	Total converted	$1,288,352
Lifetime income tax, doing nothing	$503,928	Break-even	Age 62, ahead from the start
Present value, tax + IRMAA	$288,349 vs $295,188	Traditional at the end	$0 vs $1,657,171

Healthcare Cost Planner

What will health insurance cost before and after Medicare?

ACA marketplace premiums and subsidies for the gap before Medicare, then Medicare Part B, Part D, Medigap and IRMAA surcharges by income, on 2026 rules.

06

retcalc.app/healthcare

ACA premium before 65

$2688/mo

Est. net Silver premium, whole household

Years on the ACA bridge

5 yrs

Until Medicare at 65

Medicare from 65

~$716–$876/mo

Estimated total, for the couple

Before 65: the ACA bridge

Your income of $95,000 is **449% of the federal poverty level** for a 2-person household — this ratio drives your subsidy. That income is above the 400% FPL cutoff, so there is no credit and you would pay the Silver plan's full list price. The enhanced credits that removed this cutoff (capping your share at 8.5% of income) expired after 2025.

2026 ACA rules

Benchmark Silver plan (IL est., ages 60 and 59) — the reference used to price your credit	$2688/mo
Premium tax credit	none (income above 400% FPL)
Your net premium	**$2688/mo**

If Congress restores the enhanced credits (in place 2021–2025)

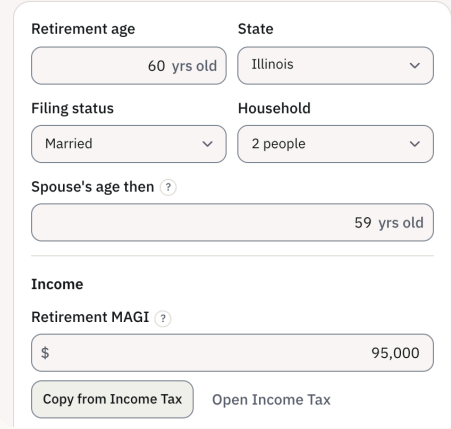
From 65: Medicare

Part B covers doctor visits, outpatient care, and preventive services. **Part D** covers prescription drugs. Most enrollees add a **Medigap supplement** (like Plan G) which covers deductibles and copays that Parts A and B leave unpaid, capping your out-of-pocket exposure. Higher incomes trigger IRMAA surcharges that raise the Part B and Part D premiums.

IRMAA Standard: standard premium

Part B premium	$203/mo per person
Part D estimate avg plan, no IRMAA	~$35/mo per person
Medigap Plan G varies by state, insurer & age	~$120–$200/mo per person
Estimated total per person	**~$358–$438/mo**
For the couple both at same IRMAA tier	~$716–$876/mo

Sample couple retiring at 60, with $95,000 of income: just over the subsidy cliff, so the full ACA premium until 65, then Medicare.



You enter

- Retirement age, status and household size
- State and income in retirement (MAGI)
- Social Security, if any, and your own quote

Why it matters

Before 65, the subsidy depends on income, so how a retiree draws money down can change the premium by thousands. The planner shows the household's share of the poverty line, warns near the subsidy cliff, and compares plan tiers. From 65 it prices Part B, Part D and Medigap, with the IRMAA tier income puts you in.

FIRE Calculator

When does work become optional?

Finds the age a portfolio reaches financial independence. Coast FIRE mode finds the moment you could stop contributing and let growth carry you the rest of the way.

07

retcalc.app/fire

Rate band | Historical — Your rate of return, and every market since 1926

FIRE age, at your rate of return

66.6

36.6 years from now

Portfolio at FIRE

$1,024,146

In today's dollars

Target portfolio

$1,000,000

Spending $40,000 per year at 4.00%

Historical success rate **50%**

1% (aggressive) — 99% (conservative)

FIRE age, in market history

Age 60.3

(30.3 years from now)

50% of historical windows since 1926 show the portfolio reaching the target by age 60.3.

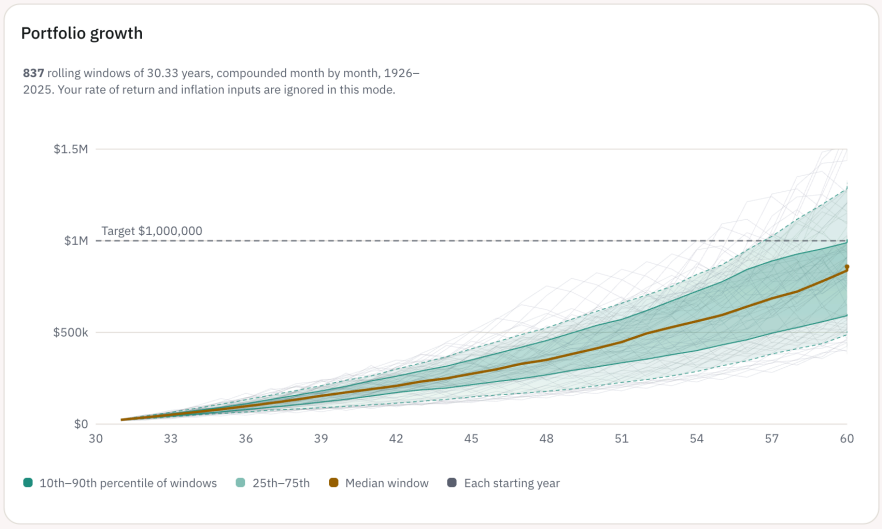
Annual withdrawal	$40,000	Nominal return	8.50%
Contributions, future dollars	$444,000	Real return	5.34%
Investment gains, future dollars	$2,603,309		

You enter

- Age, savings and monthly saving
- Return and inflation
- Spending and withdrawal rate
- FIRE or Coast FIRE

Historical mode

Runs every rolling window since 1926. A success-rate slider then gives a second answer: the age reached in, say, 90% of history.



Portfolio growth, every historical window since 1926.

Portfolio Backtest

What would this mix have actually done?

Pick a mix of stocks, small-cap value, bonds and cash, then see what it actually did: return, volatility, the worst year, the deepest fall, and every rolling window back to 1926.

08

retcalc.app/backtest

Return, per year

9.65%

Compound annual growth, 1926–2025

After inflation	Volatility	Deepest fall
6.52%	15.26%	-50.87%
Inflation averaged 2.94% per year	Standard deviation of annual returns	1928–1932

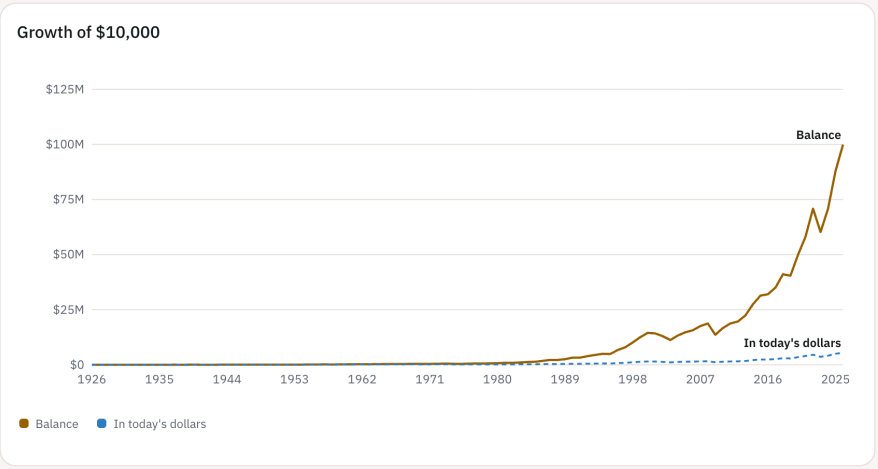
Best year	↑ 44.54% in 1933	Grew to	$99,910,217

You choose

- Stocks, small-cap value, bonds and cash
- Rebalancing yearly, every few years, or by band
- Any span from 1926 to 2025, with one-tap eras

You see

- Return per year, before and after inflation
- Volatility and the deepest fall
- Growth of $10,000
- Rolling returns and how prices multiplied
- Every year in a table



Rolling returns

nominal

Nominal | Real | CSV

Window	Periods	Worst	Median	Best	Positive
1 year	100	↓ -33.24%	13.65%	↑ 44.54%	74%
5 years	96	↓ -8.47%	10.58%	↑ 24.14%	94%
10 years	91	↑ 0.85%	9.61%	↑ 17.22%	100%

Rolling returns. The best, worst and typical result for every holding period, from every start since 1926.

College Savings

How much should I save for college?

How much to save each month to cover tuition, with presets for a public in-state school, a private college and an elite one, and room for several children sharing one account.

09

retcalc.app/college

Save per month

$558

for 18 years at 6.0%

Total projected cost

$232,269

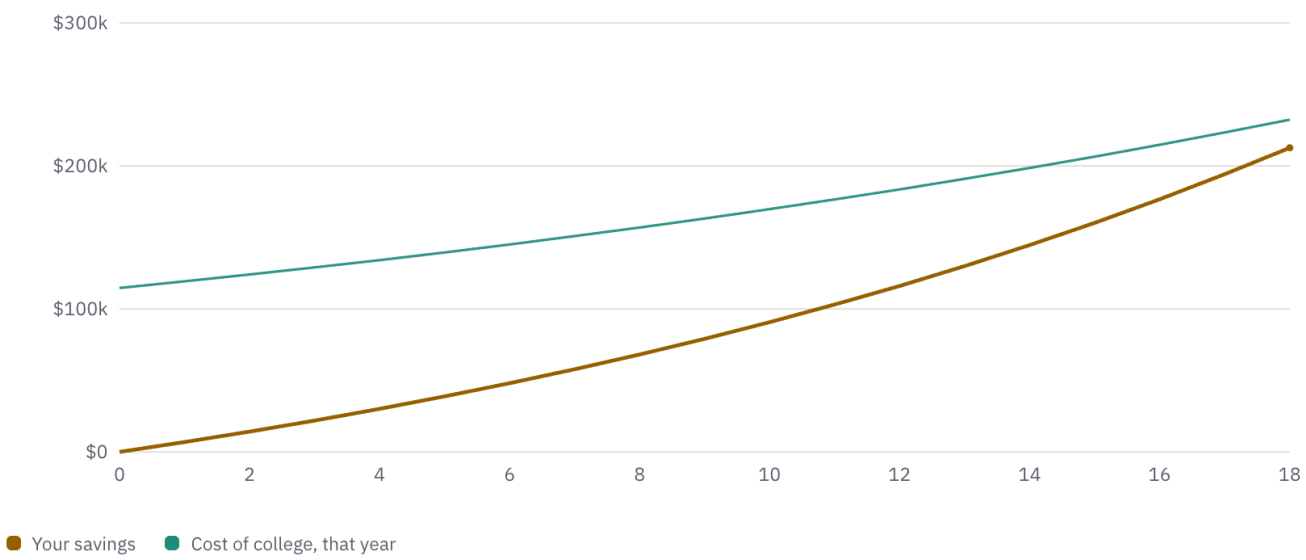
4 years at $27,000/yr, 4.0% inflation

Needed when college starts ?

$212,674

Less than the $232,269 total, since what's saved keeps earning 6.0% while later years' tuition is paid

Savings over time



Savings over time against the cost when college starts. Sample: one child, 18 years away, public in-state preset.

Presets, per year

- Public in-state, about $27,000
- Private non-profit, about $59,000
- Elite, about $82,000
- Or a custom figure

You enter

- Years until college, and years in it
- What is already saved
- Return and tuition inflation
- Each child, if there is more than one

You get

- The monthly saving needed, stepping down for younger children
- The total projected cost
- What is needed today, or when college starts

Source

College Board, Trends in College Pricing, updated each year.

Rent vs. Buy

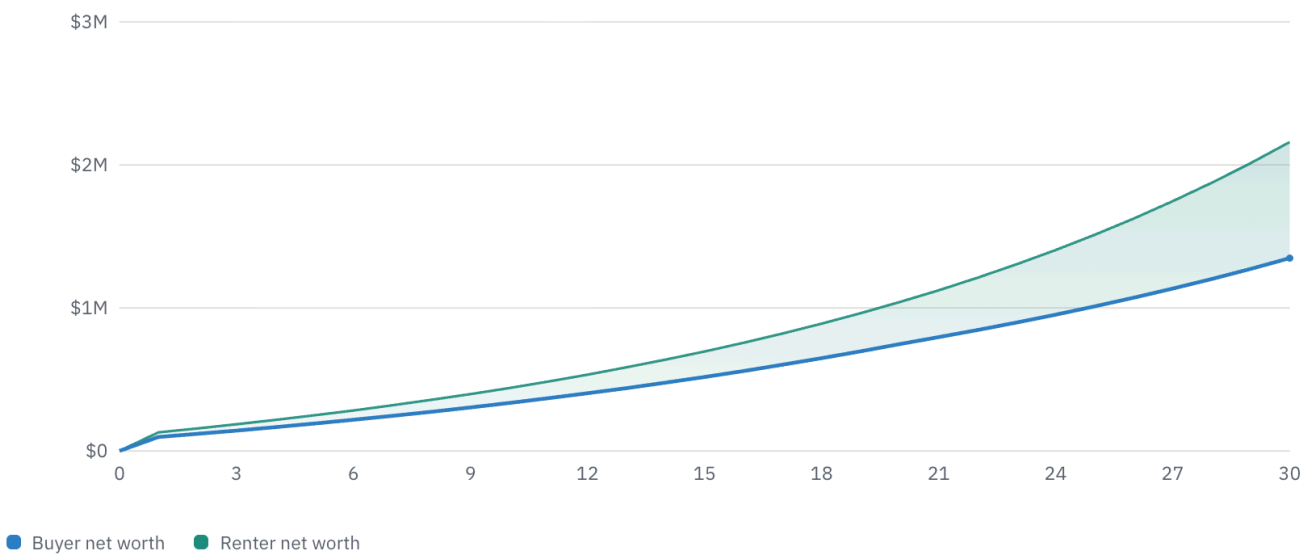
Should I rent or buy?

Compares a buyer's and a renter's net worth over any horizon, counting what the down payment could have earned invested and what the home gains in value.

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retcalc.app/rentbuy

Net worth over time



Better after 30 years

Renting

Ahead by $811,712

Break-even

Renting stays ahead throughout

Buyer net worth ?

$1,347,907

Renter net worth ?

$2,159,618

Each month, to start		Up front	
Buying, all in	$3,393/mo	Loan amount	$360,000

Each month, to start: the buyer's costs beside the renter's, and which side is ahead.

It counts

- Price, down payment, rate and term
- Property tax, insurance and maintenance
- Closing and selling costs

And

- Rent, and how fast it rises
- Home appreciation and investment return
- Tax on gains

The verdict

Which side is ahead after the chosen number of years, by how much, and the year the lines cross.

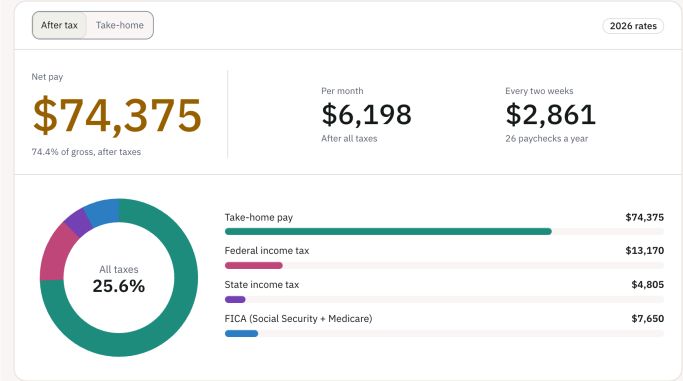
Income Tax

How much tax will I owe?

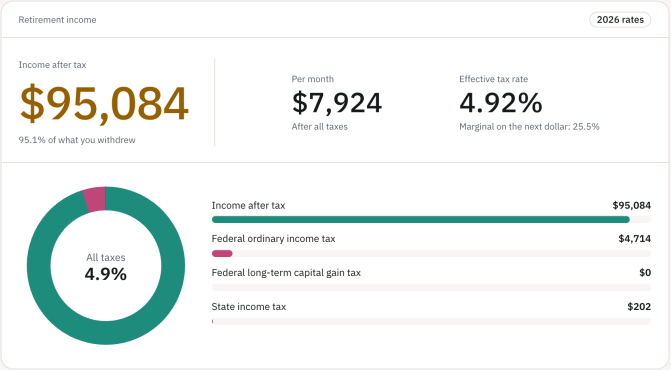
2026 federal, state and FICA tax on a salary, or a retirement mode that shows what a year of withdrawals costs across traditional, Roth, brokerage and Social Security.

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Working. $100,000 salary, single, Illinois.



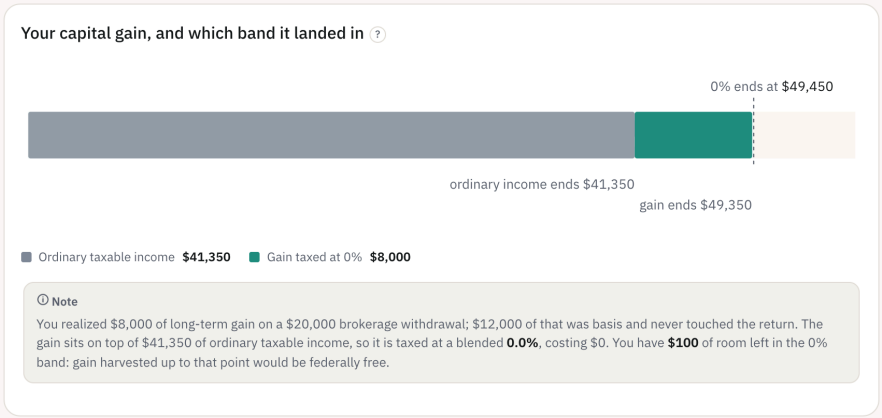
Retired. $100,000 from four sources, 65+.

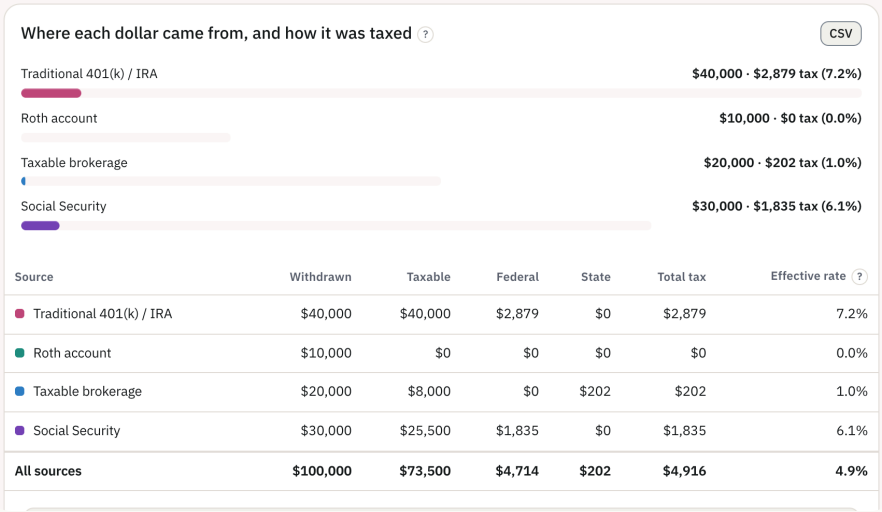
Covers

- Federal brackets, standard or itemized deduction
- Social Security tax per earner up to the $184,500 wage base, Medicare and the 0.9% surtax
- All 50 states plus DC, with their retirement-income rules
- Pre-tax deductions by kind and by earner: 401(k), IRA, HSA, premiums

Retirement mode adds

- Long-term gains stacked into the 0/15/20% bands
- The 3.8% net investment income tax
- Taxable Social Security (the 0/50/85% tiers)
- The 2026 senior deduction
- A true marginal rate, from rerunning the whole return with $1,000 more





Source	Withdrawn	Taxable	Federal	State	Total tax	Effective rate
Traditional 401(k) / IRA	$40,000	$40,000	$2,879	$0	$2,879	7.2%
Roth account	$10,000	$0	$0	$0	$0	0.0%
Taxable brokerage	$20,000	$8,000	$0	$202	$202	1.0%
Social Security	$30,000	$25,500	$1,835	$0	$1,835	6.1%
All sources	**$100,000**	**$73,500**	**$4,714**	**$202**	**$4,916**	**4.9%**

Budget

Where does my money go?

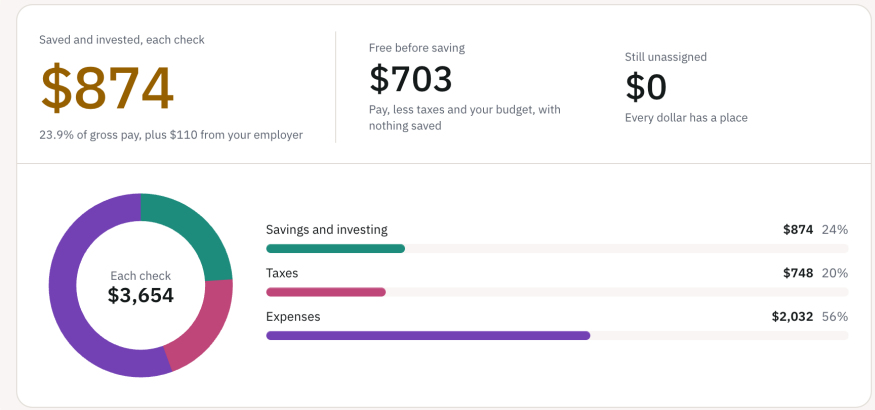
Lay out the expenses, then see how each paycheck covers them and where the rest goes. Two tabs: a Budget, and a Paycheck that turns gross pay into taxes, bills and savings, line by line.

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retcalc.app/budget



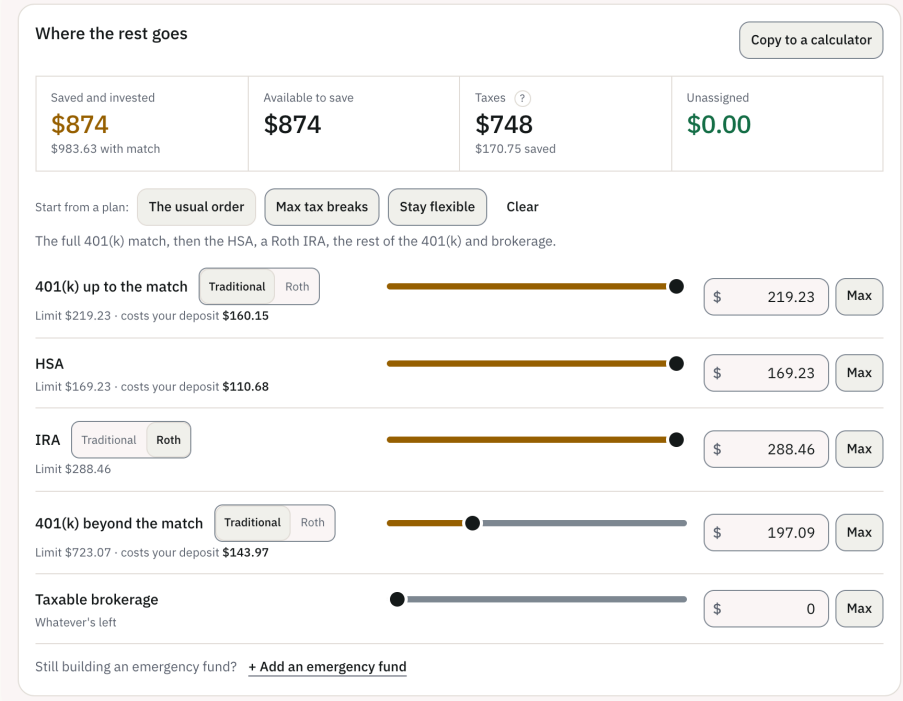
Budget tab. 18 lines in six groups, monthly or yearly.



Paycheck tab. Each check split into saving, taxes and expenses.

Three one-click plans

- **The usual order:** match, emergency fund, HSA, Roth IRA, more 401(k), brokerage
- **Max tax breaks:** everything pre-tax first
- **Stay flexible:** money reachable before retirement



Where the rest goes. Six lines, each capped at its 2026 limit; taxes rerun as the sliders move.

Two incomes

Married with two incomes plans by the month, taxes the couple jointly, and gives each spouse their own pay schedule, 401(k) match and HSA limit.

Connected

Opens in Income Tax; sends the savings split, by account type, to the Advanced or Stages calculator.

Debt Payoff

Which debt should I pay off first?

List what you owe, then watch avalanche and snowball race each other: payoff date, total interest, and exactly what ordering by motivation instead of rate costs.

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retcalc.app/debt

Debt-free

Sep 2030

3y 11m from now

Total interest

$8,155

on $49,400 borrowed, $57,555 paid in all

Saved vs. minimums

$10,167

6y 4m sooner

You get

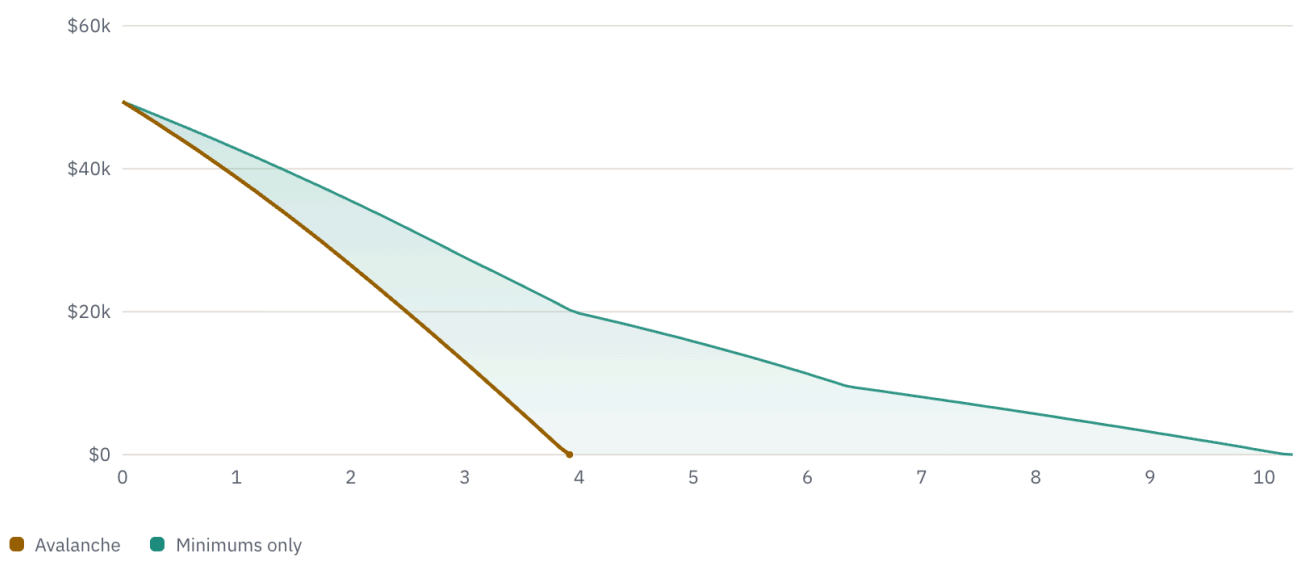
- The debt-free date and total interest
- What it saves against paying minimums
- Avalanche against snowball, with a verdict
- The payoff order
- A blended rate, minimums that don't cover interest, and what $100 more per month would do

Avalanche vs. snowball CSV

Avalanche costs $424 less in interest and finishes 1 month sooner. Snowball clears your first debt 1y 5m earlier. That early win is the whole argument for it, and here it costs $424. Your blended rate is 9.7% across 4 debts, and you're putting $1,230 per month at it. Another $100 per month would clear it 4 months sooner and save $929 more.

Approach	Debt-free	How long	Total interest	First debt gone
Avalanche, highest rate first ✓	**Sep 2030**	**3y 11m**	**$8,155**	**1y 11m**
Snowball, smallest balance first	Oct 2030	4 years	$8,578	6 months
Minimums only, no extra	Jan 2037	10y 3m	$18,321	3y 1m

What you owe, month by month



What you owe, month by month. Sample: a credit card, a store card, a car loan and a student loan, with $300 extra per month.

Mortgage Calculator

What will the house cost each month?

The full monthly payment, how the balance falls, and amortization year by year, with extra payments, a recast or a refinance to compare against the loan as it stands.

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Extra payments and refinancing

Extra payments, recasting, or refinancing

Yes, show these options

Paying extra each month or once, recasting after a lump sum, and refinancing, each with its before and after.

Paying extra

Extra toward principal

$ 300 /mo

One-time extra payment

$ 10,000

In month

24

After that payment

Keep the same payment, finish early

Refinancing

New rate

6.25 %

New length

30 years

Closing costs

$ 4,500

With the extra payments

Payoff
year 20, month 5
9y 7m sooner

Interest saved
$201,179

Refinancing to 6.25% for 30 years

New payment
$2,217/mo
Principal & interest, from $2,456/mo

Monthly change
Saves −$239/mo

Breaks even on closing costs
1y 7m

Over the life of the loan
Saves $39,855

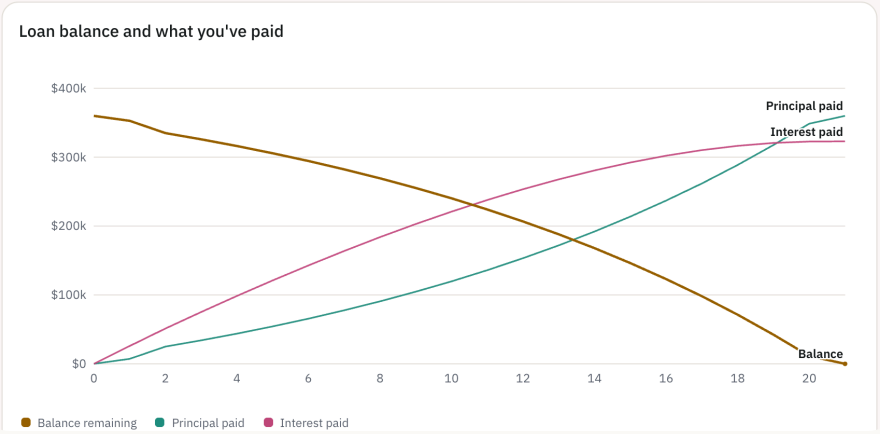
Sample: $450,000 at 7.25%, plus $300 per month, $10,000 in month 24, and a refinance to 6.25%.

What-ifs

- Extra every month, or a lump sum
- Recast, or keep the payment and finish early
- Refinance, with the break-even

Defaults

The rate starts at Freddie Mac's weekly survey, checked each quarter.



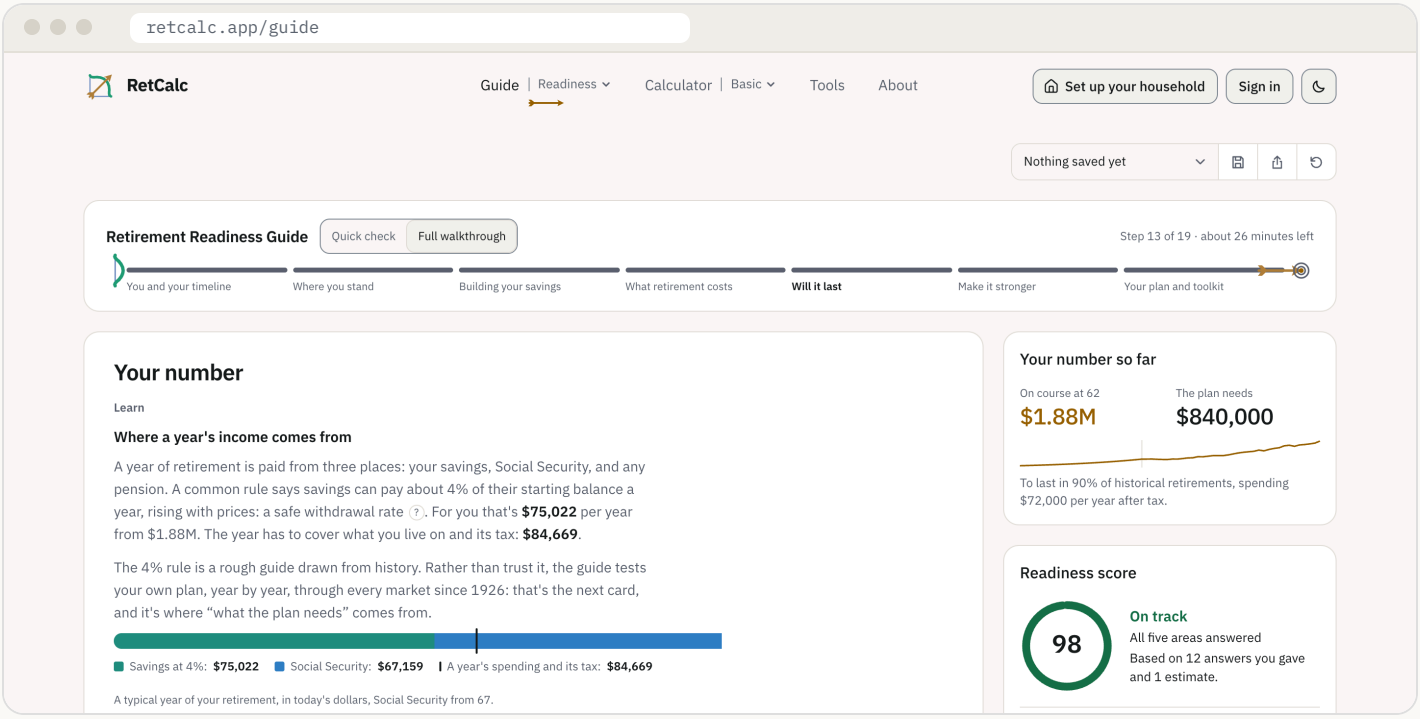
Retirement Readiness Guide

How ready am I, and what should I do next?

A guided check-up. It finds the number to aim for, tests the plan against every market since 1926, scores readiness out of 100 and lists the moves that matter, in the order that matters.

G1

retcalc.app/guide



Two paces

QUICK CHECK · ABOUT 10 MIN

Twelve short cards, three questions at most on each. The number by the sixth card, then a test of whether it lasts. Defaults stand in for the details, and every one says so.

FULL WALKTHROUGH · AN HOUR OR MORE

Every card, with trips into the tools that find the numbers you don't know: children, raises, a paid-off mortgage. It saves as you go.

Seven chapters

01 You and your timeline
02 Where you stand
03 Building your savings
04 What retirement costs
05 Will it last
06 Make it stronger
07 Your plan and toolkit

The readiness score

Retirement outlook 40
Full marks when the plan lasted in 95% of history

Savings rate 20
15% of gross income, or a plan that already lasts

Emergency fund 15
Six months of spending in cash

Debt 15
Nothing owed at 8% or more, besides a mortgage

Monthly cash flow 10
Spend no more than 80% of take-home

On track at 85 and up, then Nearly there, Getting there, Needs work and Needs attention. A confidence line says how many answers were entered, estimated or defaulted.

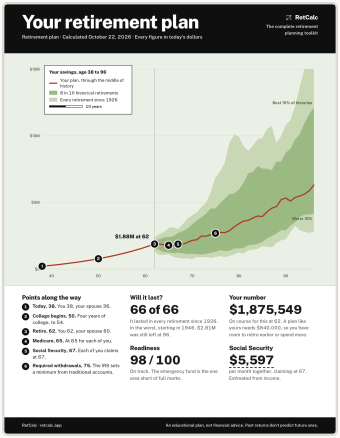
The plan, on paper

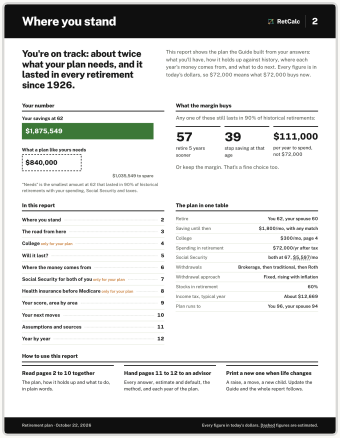
A full-color report in your own numbers.

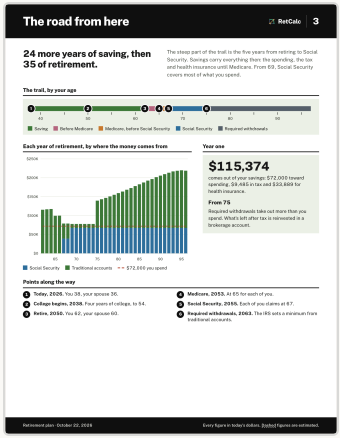
The Guide ends in a report of about a dozen US Letter pages, built in the browser and saved as a PDF: what you will have, how it holds up against every market since 1926, Social Security, taxes and health insurance, and the next moves. Below, the report for a sample couple, 38 and 36, retiring at 62.

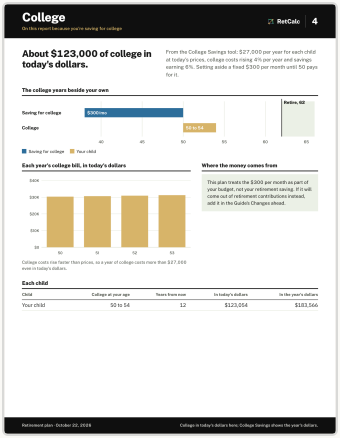
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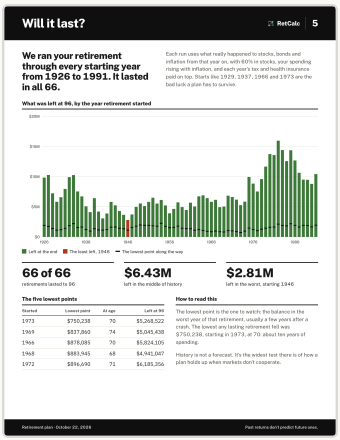
retcalc.app/guide/report

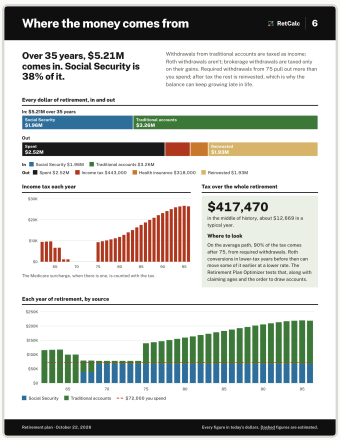


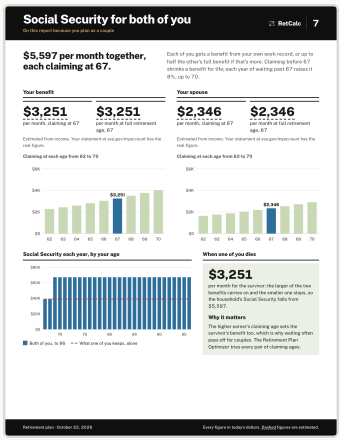




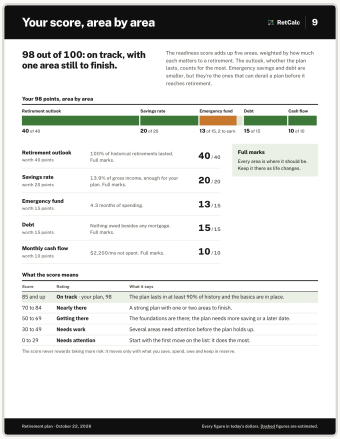


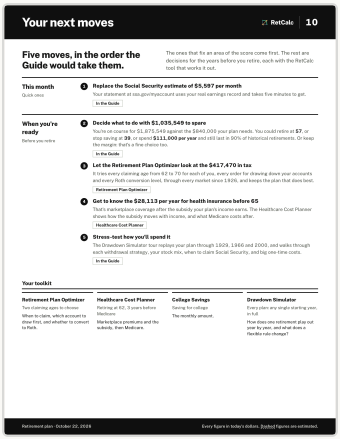


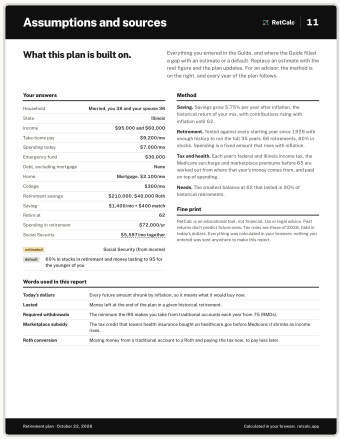


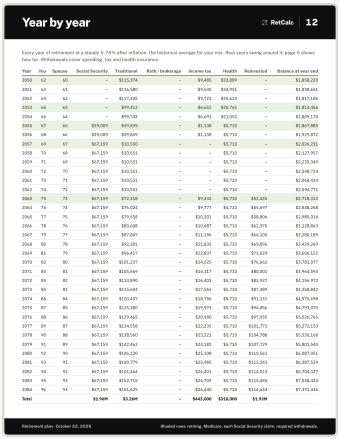












Paycheck to Retirement

Where should each paycheck go?

For anyone who would rather get straight to the numbers. Five steps, about fifteen minutes: take-home pay after tax, a budget, where each paycheck goes, and where that saving leads by retirement, children included.

G2

retcalc.app/guide/paycheck

START
Start
Ages, filing status, pay and state

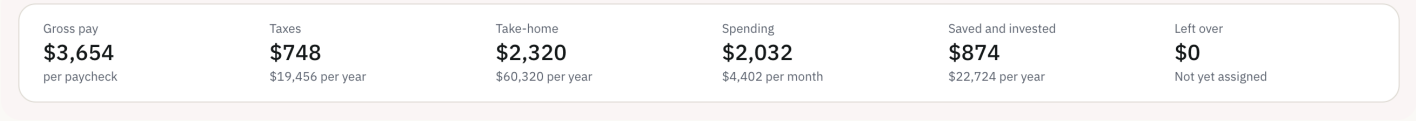
STEP 1
Budget
Spending, line by line

STEP 2
Paycheck
Tax, then 401(k), HSA, IRA, brokerage

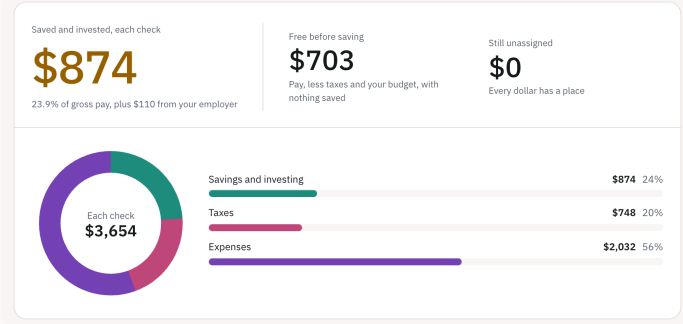
STEP 3
Your plan
The Advanced calculator, seeded

STEP 4
Life stages
Optional: Stages, with children

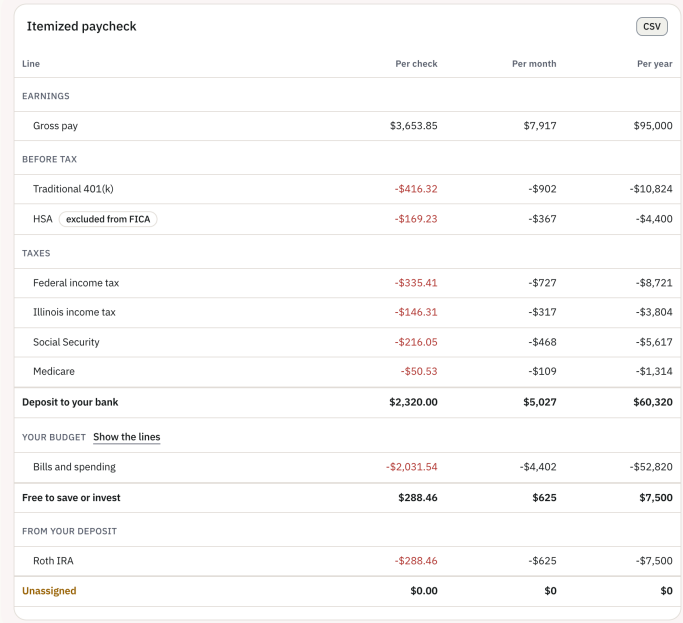
STEP 5
Summary
Copy into my tools



The running totals stay pinned above every step: pay per check, taxes, spending, saving and what is left.



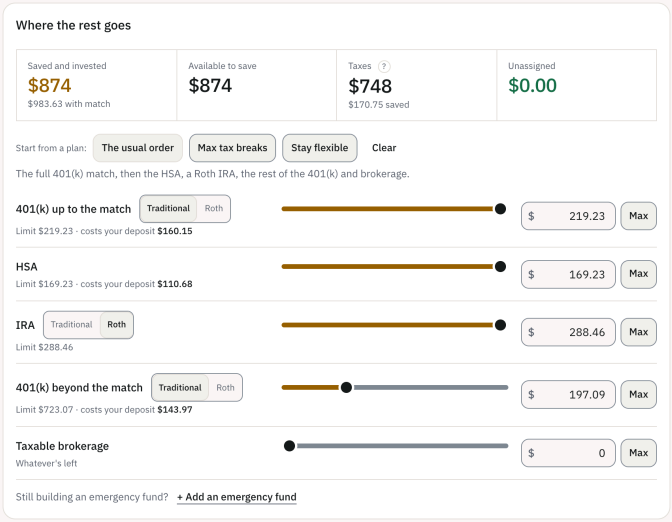
Step 2, Paycheck. Each check, split.



Itemized paycheck — CSV

Line	Per check	Per month	Per year
EARNINGS			
Gross pay	$3,653.85	$7,917	$95,000
BEFORE TAX			
Traditional 401(k)	-$416.32	-$902	-$10,824
HSA excluded from FICA	-$169.23	-$367	-$4,400
TAXES			
Federal income tax	-$335.41	-$727	-$8,721
Illinois income tax	-$146.31	-$317	-$3,804
Social Security	-$216.05	-$468	-$5,617
Medicare	-$50.53	-$109	-$1,314
Deposit to your bank	**$2,320.00**	**$5,027**	**$60,320**
YOUR BUDGET Show the lines			
Bills and spending	-$2,031.54	-$4,402	-$52,820
Free to save or invest	**$288.46**	**$625**	**$7,500**
FROM YOUR DEPOSIT			
Roth IRA	-$288.46	-$625	-$7,500
Unassigned	$0.00	$0	$0

The itemized paycheck, per check, per month and per year.



Where the rest goes. Pick a plan or move the sliders.

It runs the real tools

Budget, Advanced and Stages run inside the walkthrough on its own copy of the numbers, so nothing in the standalone tools changes until **Copy into my tools** sends the results to Income Tax, Budget, Advanced and Stages.

It ends in a report

Cover, the paycheck, the spending, the saving, the road to retirement, life stages if used, and the assumptions.

ONE HOUSEHOLD, START TO FINISH

From a paycheck to a retirement that lasts.

A sample couple, carried through four tools by the tools' own buttons. Nothing was typed twice: each tool handed its answer to the next. Every figure on these two pages is from the live site.

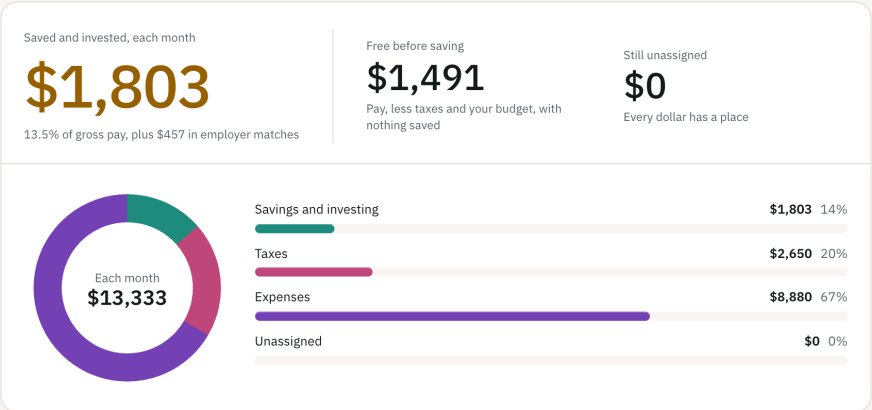
34 and 32	$160,000	$60,000	2	65
Alex and Jordan, in Illinois	$92,000 and $68,000 pay, per year	saved so far	children planned, the first in two years	when Alex retires, 31 years out

1 BUDGET

Where each paycheck goes

Their budget goes in line by line, then both paychecks on the Paycheck tab. One click on **The usual order** fills both 401(k) matches, an HSA, two Roth IRAs and more 401(k), and reruns their taxes as it goes.

They can save **$1,803 per month**, 13.5% of pay, plus $457 per month from their employers.



Copy to a calculator → Stages carries **$832.35 every two weeks**, split by account type, with the salaries and the match.

2 STAGES

Saving around the children

The **Planning for kids** builder turns one 31-year stage into nine, from *Before children* to *Empty nest*. Childcare and the teen years take the saving to nothing for 18 years; it comes back as the children leave.

At 65 that is **$743,576 in today's dollars**, from $487,735 put in over 31 years.

Your stages

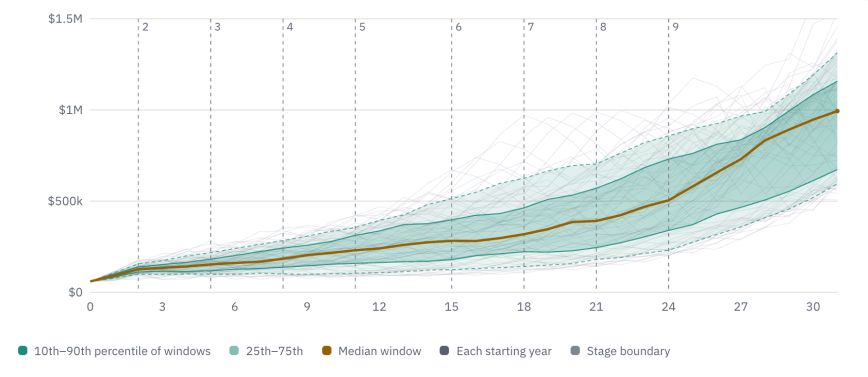
1 2 3 4 5 6 7 8 9

Year 0 — 31 yrs in all

1	Before children	2 yrs	$832/2wk
2	A child in childcare	3 yrs	$0/2wk
3	Two children, both in childcare	3 yrs	$0/2wk
4	Two children, one in childcare	3 yrs	$0/2wk
5	School years	4 yrs	$0/2wk
6	School years, one a teenager	3 yrs	$0/2wk
7	The teen years	3 yrs	$0/2wk
8	A teenager	3 yrs	$278/2wk
9	Empty nest	7 yrs	$832/2wk

STEP 2, TESTED AGAINST HISTORY

The same plan run through all 829 31-year windows since 1926, with a dashed line where each stage begins. The median finish is **$993,511**; the worst window, starting in 1951, still reaches $488,459.



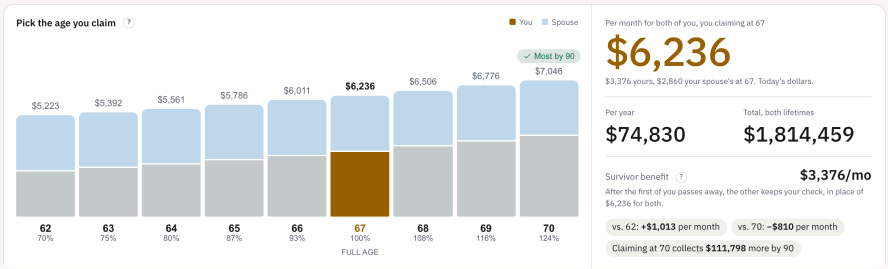
↓ `Test withdrawals → Drawdown Simulator` carries the portfolio: **$743,576**, the balance in today's dollars.

3 SOCIAL SECURITY CALCULATOR

When to claim

Born in 1992 and 1994, with the same pay. Claiming at 67, their full retirement age, brings **$6,236 per month** together: $3,376 for Alex and $2,860 for Jordan, in today's dollars.

Waiting until 70 would add $810 per month and more over a long life; the calculator shows by how much at every age.

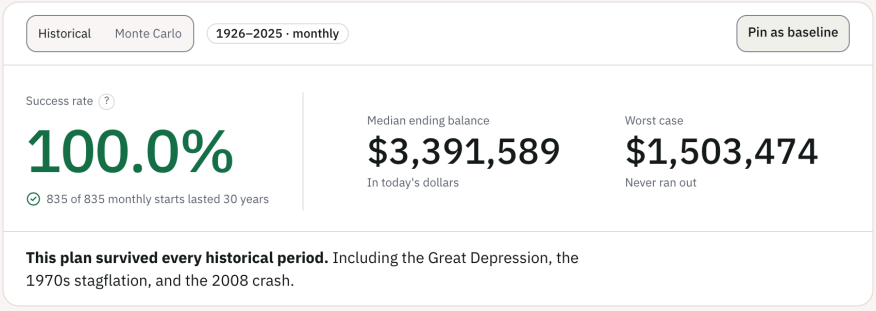


↓ `Copy from Social Security Calculator` carries both benefits into the Drawdown Simulator: **$40,506 and $34,323 per year**, starting at 67.

4 DRAWDOWN SIMULATOR

Will it last?

From 65, the 4% rule draws **$29,743 per year**, rising with inflation, and Social Security joins at 67. The plan is replayed through every monthly start since 1926, 835 of them, for 30 years each.



WHERE ALEX AND JORDAN LAND

$104,573
per year in retirement, in today's dollars: $29,743 from savings and $74,830 from Social Security

835 of 835
historical retirements lasted the full 30 years, the Depression and the 1970s included

$1.5M
left at the end in the worst of them, in today's dollars

A sample household, not advice. Returns before retirement are the plan's own assumptions; the historical tests use real monthly returns from 1926 to 2025.

PART 3

Under the hood.

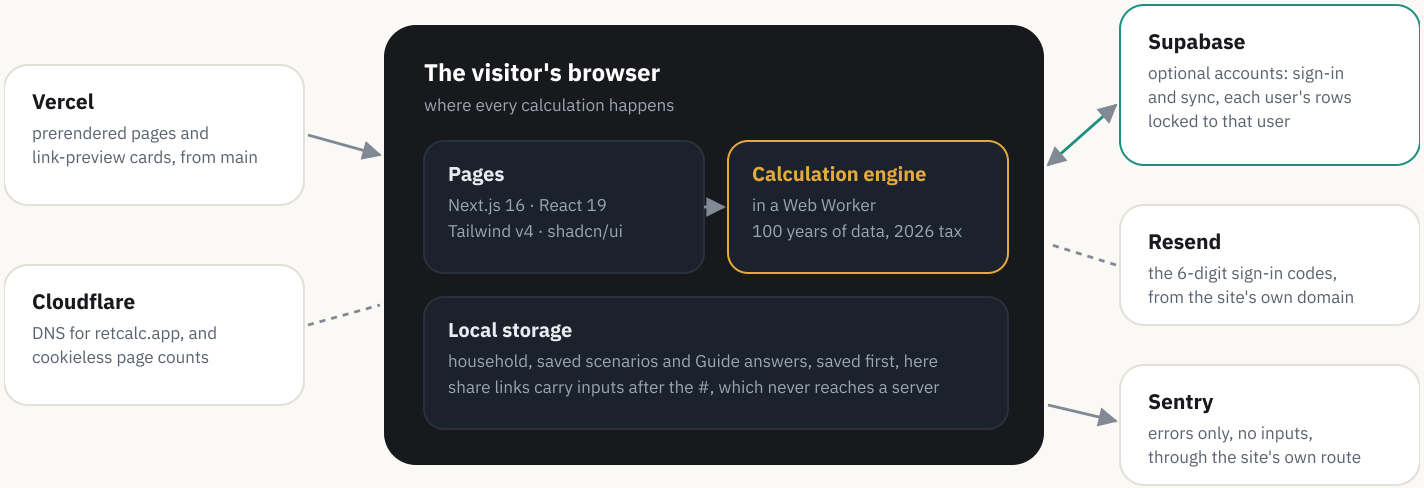
Next.js on Vercel, an engine that runs in the browser, accounts that only sync, and figures that each carry a source and a due date.

The stack	32
The calculation engine	33
Hosting and delivery	34
Accounts and privacy	35
Where the numbers come from	36

THE STACK

A static site that does its thinking in the browser.

Pages are prerendered and served from a CDN. Every number on them is worked out on the visitor's own machine, by a calculation engine that runs off the main thread. Accounts, when someone wants one, only hold their saved inputs.



Solid lines carry data. Dashed lines are supporting services. There are no API routes: the only thing the site's own server receives is error reports, with every input stripped out.

LAYER	WHAT IT USES
Framework	Next.js 16 (App Router), React 19, TypeScript 5
Look	Tailwind CSS v4 and shadcn/ui on Base UI primitives, Lucide icons; IBM Plex Sans, served by the site itself
Engine	Plain JavaScript, type-checked through JSDoc, behind typed TypeScript bridges
Checks	ESLint with zero warnings allowed, `tsc`, @shadcn/lint, a calculation suite, unit and end-to-end tests
Design	A written design system: Graphite (dark, the default) and Paper (light) themes, one amber accent, gain and loss colors, five chart hues

BY THE NUMBERS

39
page routes

~55,600
lines of app code

~7,400
lines in the engine and its data

29
explanatory articles

The design rules are enforced, not just written: the linter rejects raw colors, one-off sizes and inline styles, and a new look has to be a component variant.

THE PALETTE, FROM THE DESIGN SYSTEM

Graphite	Surface	Raised	Amber
#0e1116	#151a21	#1c222b	#e3a93b

Paper	Raised	Ink	Amber
#faf9f6	#f3f1ec	#17191c	#9a6200

Chart series Gain and loss

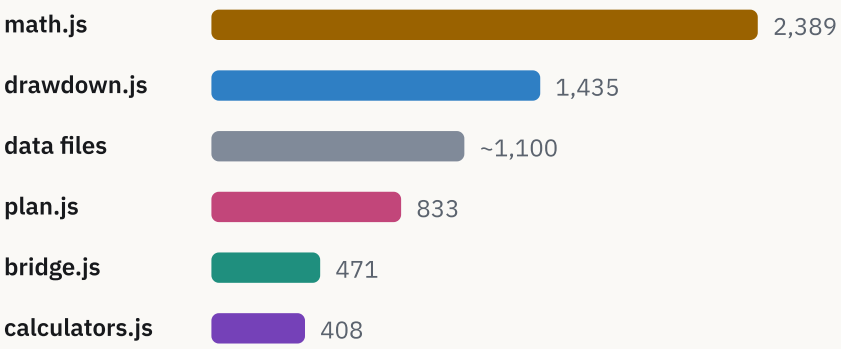
Typeface: **IBM Plex Sans**

THE CALCULATION ENGINE

About 5,700 lines of finance, and a test for every claim.

One engine serves every page: projections, Monte Carlo, federal and state income tax, FICA, Social Security, RMDs, Roth conversions, ACA subsidies, Medicare and IRMAA, mortgages, debt, rent against buy, college, FIRE, the 72(t) bridge and historical withdrawal backtests.

THE ENGINE, BY FILE (LINES)



Same seed, same chart

Every simulation shares one seeded random number generator (mulberry32), so a chart doesn't jump when an unrelated input changes. A **Re-roll** button moves the seed for all of them at once.

Off the main thread

A build step bundles the engine, its data and the Guide's planning jobs into one Web Worker, so a 300,000-retirement Optimizer run doesn't freeze the page.

THREE WAYS TO ASK "WHAT IF MARKETS DON'T COOPERATE?"

Historical replay	Every starting month (or January) since 1926, run forward through what actually happened. The Drawdown Simulator's 30-year test has 835 of them.
Monte Carlo, calculators	5,000 lognormal runs per redraw, set so the average matches the rate entered: with zero volatility it draws the plain projection exactly.
Monte Carlo, drawdown	A block bootstrap of real history from 1927. Whole runs of consecutive years are drawn together, so the 1930s and 1970s keep their streaks.

HOW A CHANGE GETS TO THE LIVE SITE

A change
to the engine, a tool's defaults, an article, page text or the tests

→

Pre-commit gate
~618 checks in 55 groups
209 of them hold the worked examples in all 29 articles to what the calculator produces
A failure blocks the commit.

→

On every push
the calculation suite again, plus numbers baselines: each tool's headline figures and seven Guide sample households
lint · typecheck · production build
unit tests · 17 end-to-end browser specs

→

Live
main deploys to Vercel; previews deploy for every branch first

~618 checks in the calculation suite, in 55 groups	**209** of them tie the 29 articles' worked examples to the engine	**7** sample households whose Guide results are pinned	**17** end-to-end browser specs	**1,200** months of market history every engine walks

HOSTING AND DELIVERY

Fast because there is almost nothing to serve.

RetCalc runs on Vercel and deploys from the main branch. Pages are prerendered when the site is built, the homepage's and tool list's own figures included, so a visit is a handful of static files and a JavaScript engine that does the rest.

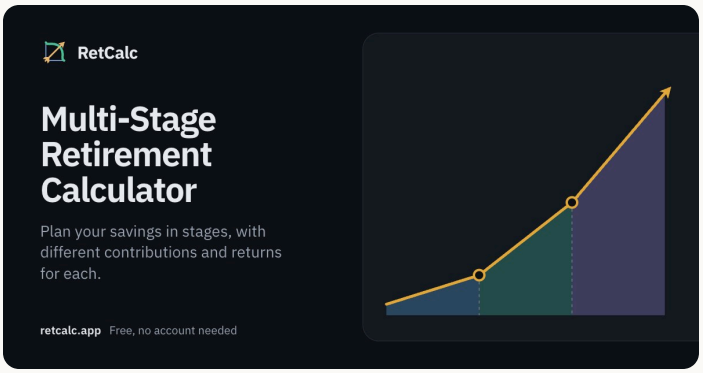
PIECE	HOW IT IS DONE
Hosting	Vercel. Production builds from main; every branch gets its own preview. Before 5 October 2026 the site was a hand-built static site; it now lives on in a private archive for comparison.
Rendering	Prerendered pages and articles. There are no API routes: all calculation happens in the browser, and the one thing the site's own server receives is error reports, stripped of inputs, which it passes on to Sentry.
Domain	retcalc.app, with DNS on Cloudflare; www redirects to the bare domain.
Analytics	Cloudflare and Vercel Web Analytics: cookieless page counts, production only. No ads and no ad trackers.
Email	Sign-in codes go out through Resend from the site's own domain. Mail to contact@retcalc.app is forwarded separately.
Installable	A web app manifest and icons, so it can be added to a phone's home screen, with a bottom tab bar there.

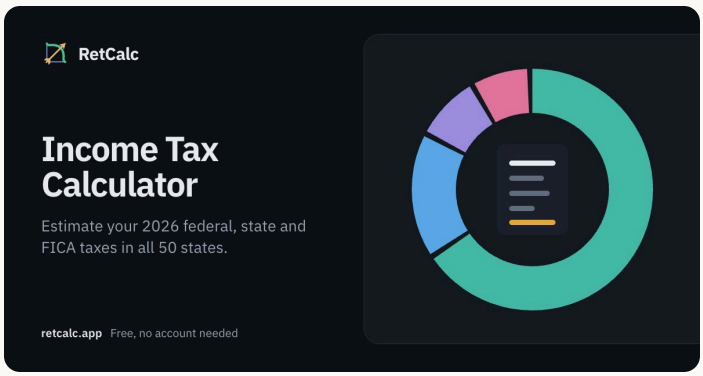
Error reports that can't leak a plan

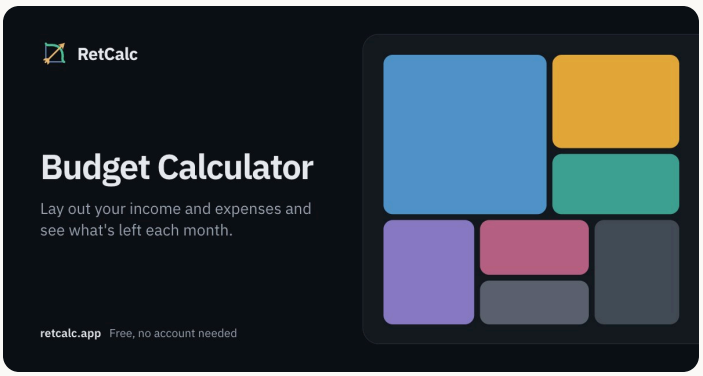
- Errors only: tracing, replay and debug code are compiled out
- Production only
- Addresses cut to the path, so the query and the # that holds a tool's inputs never leave the page
- No console breadcrumbs, user details or local variables
- Sent through the site's own route, so ad blockers don't drop them

LINK PREVIEWS, DRAWN FOR EVERY PAGE







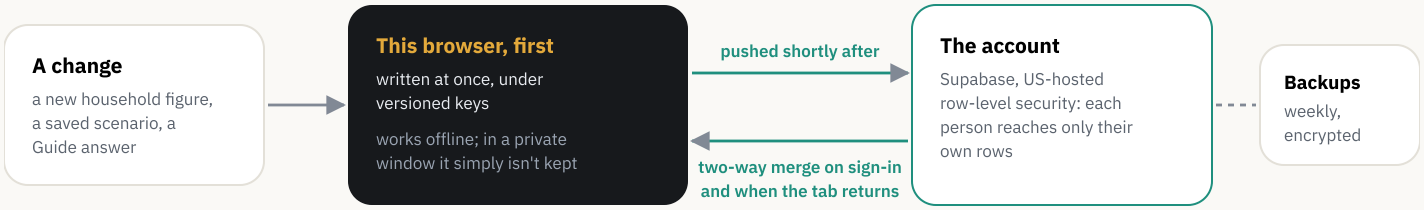


Each page has its own 1200×630 card: a title, a line of text and one of about 21 illustration types. A development-only page draws them and a headless browser saves them as static images, so sharing a link to any tool shows that tool.

ACCOUNTS AND PRIVACY

Local first. An account is only a way to sync.

Everything works without signing in. A signed-in visitor gets the same site, with their household, saved scenarios and Guide kept in step across devices. Saving never waits on the network, and a sync never throws anything away.



If both sides changed a scenario with the same name, both are kept. If two different Guides meet, one becomes a saved plan. The merge is pure functions, unit-tested.

WHAT AN ACCOUNT STORES

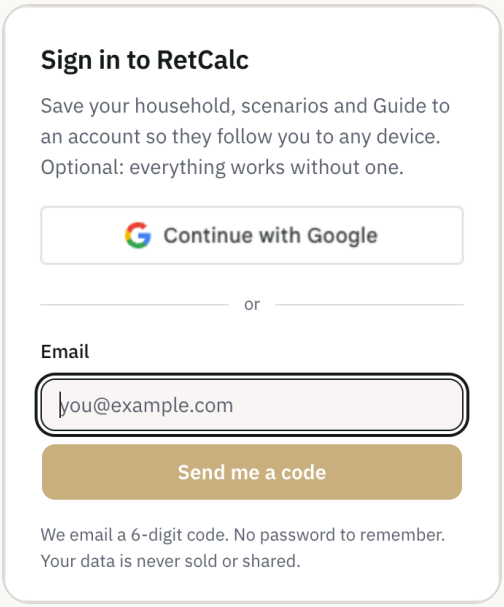
- Email, and a first name if given
- The household: ages, retirement age, state, savings, monthly saving, income, spending
- Saved scenarios, by tool
- Guide answers and plan, and the Paycheck plan
- When each was last changed

WHAT IT NEVER STORES

- Numbers typed into a tool but not saved (they stay in the tab, and survive a reload)
- Browsing activity
- Anything for advertisers: there are none, and no data is sold or shared
- Share-link inputs, which ride after the # and never reach a server

SIGNING IN, AND OUT FOR GOOD

- A 6-digit code by email, or Continue with Google (whose script loads only when the sign-in box opens)
- Download everything, any time
- Delete the account: every session is signed out first, then the account and its synced data are deleted. It can't be undone



Signing in, as it looks on the site.

No password to remember: a code by email, or Google. The dialog says up front that an account is optional and that data is never sold or shared.

The privacy page names every service that touches data (Supabase, Google, Resend, Vercel, Cloudflare and Sentry) and what each one sees. A dedicated test script checks the row-level security rules.

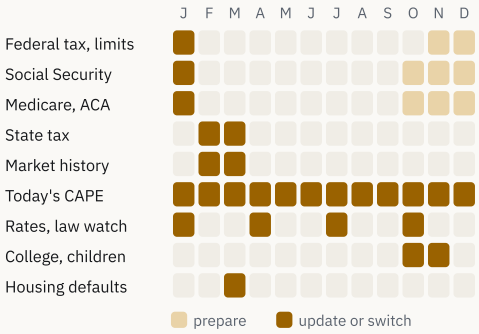
WHERE THE NUMBERS COME FROM

22 figures, each with a source and a due date.

Tax rules, Medicare and ACA prices, market history and sensible defaults all go stale. Each one is tracked in a list with its source, how often it changes, when it was last updated and when it is next due. Anything coming due is flagged at the start of every working session.

FIGURE	SOURCE	HOW OFTEN
TAXES		
Federal brackets, deductions	IRS Revenue Procedure (2025-32 for 2026)	Yearly
Contribution limits	IRS notices, incl. HSA limits	Yearly
Social Security figures	SSA's October fact sheet	Yearly
State income tax, 50 + DC	Tax Foundation's January survey, state revenue departments	Yearly
State retirement rules	State revenue departments, checked against Tax Foundation and AARP	Yearly
RMD tables	IRS Publication 590-B	Yearly check
HEALTHCARE		
Medicare Part B, IRMAA	CMS fact sheet	Yearly
Part D, Medigap	CMS; KFF and insurer surveys	Yearly
ACA premiums, subsidies	KFF benchmark premiums, HHS poverty guidelines, IRS percentages	Yearly
MARKETS		
Stocks, bonds, inflation	Robert Shiller's data (Yale) via multpl; BLS CPI-U	Yearly
T-bills, small-cap value	Kenneth French Data Library (Dartmouth)	Yearly
Today's CAPE	multpl.com, Shiller PE	Monthly
HOUSING AND OTHER DEFAULTS		
Mortgage rate	Freddie Mac weekly survey	Quarterly
Home price and rent	NAR and Census, Zillow, Tax Foundation	Yearly
College and child costs	College Board; USDA, CPI-adjusted; Dept. of Labor childcare data	Yearly
72(t) rate, debt rates	IRS federal rates; Federal Reserve G.19	Quarterly, yearly

THE YEAR IN UPKEEP



THE RULES FOR EVERY UPDATE

- Figures are never invented. Each is cited in a comment beside it.
- A new tax year is added beside the old one, never over it. Income Tax lets you pick the year.
- After an update the whole check suite runs, and every expected move is reviewed before baselines are re-recorded.
- Articles are re-reviewed every 90 days and show the month.

RetCalc

Find out where you stand.

Free, private, and tested against every market since 1926.

retcalc.app

RetCalc is an educational tool, not financial, tax or legal advice. Past returns don't predict future ones, and tax figures are estimates.
Screenshots are from the live site on 22 October 2026 with sample households. Contact: contact@retcalc.app